



Module Manual

Master of Science (M.Sc.)

Environmental Engineering

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Program description

Content

Environmental engineering has never been more relevant than it is today. In the past 20 years, the field has moved from purely focusing on the technical and logistical side of waste disposal to encompass material recovery and circular economy. Innovative materials, integrated material and process flow analysis, as well as the involvement of energy sector issues, have brought environmental technology - once a niche sector perceived to be partly driven by ideology - into mainstream areas of the international economy. Germany is a global market leader in many areas of environmental engineering expertise. This status requires ongoing success at different levels: innovative and integrated technology, favourable legal and economic standards and, not least, high-quality German university education in environmental sciences.

With this in mind, the International Master Program in Environmental Engineering at the Hamburg University of Technology (TUHH) focuses on current developments in environmental technology, while also providing a solid grounding in the subject's scientific and economic foundations. Students can specialise in one of three areas: (i) Environment and Climate, (ii) Water and Water Engineering and (iii) Energy and Resources.

Interdisciplinary considerations are essential to all subject areas. How can environmental pollution be reduced and valuable resources recovered at the same time? How does one measure the sustainability of a product or service? Which innovative technologies assure minimum energy use in production processes? Which environmental law constraints favour sustainable development? All these questions are relevant in the Environmental Engineering program.

Graduates of the Environmental Engineering program have a detailed understanding of key areas of environmental sciences. At the start of the Master's program, all students take compulsory courses in environmental management, waste and wastewater treatment, fluid dynamics and hydrology, and environmental analytics. In the second semester, students can choose from a number of potential core areas. These include courses e. g. in geochemical engineering, water and wastewater treatment technology. From the third semester, students develop a specialisation in one of the areas mentioned above (Environment & Climate Change, Water and Energy & Resources). In addition to course-specific modules, students also take non-technical classes in subjects such as business economics or foreign languages.

Career prospects

Graduates of the International Master Program in Environmental Engineering are sought after in a broad range of different fields and have excellent prospects in terms of career development. Graduates may work, for example, for environmental authorities, water and wastewater companies, energy and waste management companies, engineering firms or in the environmental technology industry. Owing to the breadth and diversity of the course, Environmental Engineering graduates are able to quickly familiarise themselves with new information, which is highly beneficial when working in interdisciplinary teams, as will often be the case. Worldwide, the environmental technology sector is growing strongly. Inadequate environmental management can have a significantly negative impact on the economic development of a region or country. In light of the above, Environmental Engineering graduates are international in their outlook and employed around the world. In addition to preparing students for demanding careers in industry, the Master's in Environmental Engineering also equips students with the necessary academic skills for pursuing their possible further specialisation at PhD level.

Learning target

Environmental Engineering graduates should have certain core skills and knowledge. These are listed below in the following categories: knowledge, skills, social skills and independence.

Knowledge:

1. Graduates are able to describe the fundamentals of environmental management and outline environmental standards, environmental economic instruments, the content of ISO 14001 and environmental performance evaluation.
2. They are able to explain the procedural fundamentals of important water and wastewater treatment techniques, biological waste treatment (aerobic and anaerobic) and relevant environmental chemicals and their analytical determination, particularly in water and wastewater analysis.
3. They can discuss hydrological and fluid mechanical models and the technical boundary conditions for sustainable water protection.
4. They are able to define the key principles of circular economy (water/waste & resources) and outline the fundamentals of business economics.
5. Depending on the specialisation they choose, graduates can demonstrate their broader understanding in the areas of water, energy & resources and environment & climate.

Skills:

1. Graduates are able to complete practical laboratory work in the area of municipal water engineering taking into consideration the procedure selection for water and wastewater treatment processes.
2. They are able to conduct specialist scientific research and geographical data processing and apply hydrological models.
3. They are able to argue and write scientifically.
4. Graduates are able to produce incisive individual presentations and coordinated team presentations, as practised in classes involving problem-based learning (PBL).
5. They are able to apply fundamental business economics methods.
6. Depending on their chosen specialisation, they have further skills in the areas of water, energy and waste, or biotechnology. For example, they are able to design membrane separation processes, conduct modelling in water technology, select technical and regional planning solutions for tasks in a biorefinery or analyse and evaluate integrated waste management solutions.

Social skills:

1. The degree program Environmental Engineering attracts students from all over the world. From the beginning of the course, students work in diverse teams, in which they are able to use their different skill sets and values productively when working on technical problems.
2. On completion of their studies, students are able to develop technical proposals, comprehensively review results and, where relevant, confirm them through peer discussion.
3. They can present technical solutions as a team.
4. They can also give constructive feedback to fellow students and integrate feedback on their own performance appropriately into their own work.

Autonomy:

1. Graduates of the Environmental Engineering program are able to conduct independent research using scientific literature; read test reports; gain knowledge from these reports and transfer it to the project at hand.
2. In consultation with teaching staff, they are able to evaluate their own learning in concrete terms and define subsequent steps for ongoing progress.
3. They can independently define research and development tasks for theoretical and experimental investigation of environmental issues and plan and carry out projects in this regard.

Program structure

The Master's program in Environmental Engineering is composed primarily of modules with six credit points (CPs). One CP equates to a student workload of 30 hours (classroom contact hours and study undertaken at home, including examination preparation). Master's students must complete 120 CPs in four semesters over a two-year period.

The modules are divided into: (i) **core qualification**, (ii) **specialisation** and (iii) **thesis**. For the **core qualification**, all students initially attend compulsory courses amounting to 42 CPs. These are primarily completed in the first and second semesters. Based on their individual interests, students take a further 18 CPs from a possible 30 CPs of elective courses. These modules are primarily completed in the second and third semesters. It is obligatory for students to take one business economics module and a module with non-technical courses (foreign language, art or cultural courses). **Specialisation** encompasses 12 CPs of obligatory courses (project work) and 18 CPs elective courses, to be selected from the study options in the specialisations Water, Waste and Energy, or Biotechnology. These modules are primarily completed in the third semester. In the fourth semester, students complete their **thesis** (30 CPs). This is preferably completed in the student's specialisation, though this is not obligatory. The third or fourth semester is most suited to students wishing to spend time abroad or on an industry placement as project and thesis work can be completed independent of lecture periods and in direct agreement with the supervising Professor.

Core Qualification

Module M0523: Business & Management

Module Responsible	Prof. Matthias Meyer
Admission Requirements	Successful completion of the modul "Foundations of Management"
Recommended Previous Knowledge	None
Educational Objectives	After taking part successfully, students have reached the following learning results
Professional Competence <i>Knowledge</i> • Students are able to find their way around selected special areas of management within the scope of business management. • Students are able to explain basic theories, categories, and models in selected special areas of business management. • Students are able to interrelate technical and management knowledge. <i>Skills</i> • Students are able to apply basic methods in selected areas of business management. • Students are able to explain and give reasons for decision proposals on practical issues in areas of business management. Personal Competence <i>Social Competence</i> • Students are able to communicate in small interdisciplinary groups and to jointly develop solutions for complex problems <i>Autonomy</i> • Students are capable of acquiring necessary knowledge independently by means of research and preparation of material.	
Workload in Hours	Depends on choice of courses
Credit points	6

Courses

Information regarding lectures and courses can be found in the corresponding module handbook published separately.

Module M0524: Non-technical Courses for Master	
Module Responsible	Dagmar Richter
Admission Requirements	None
Recommended Previous Knowledge	None
Educational Objectives	After taking part successfully, students have reached the following learning results
Professional Competence <i>Knowledge</i>	The Nontechnical Academic Programms (NTA) imparts skills that, in view of the TUHH's training profile, professional engineering studies require but are not able to cover fully. Self-reliance, self-management, collaboration and professional and personnel management competences. The department implements these training objectives in its teaching architecture, in its teaching and learning arrangements, in teaching areas and by means of teaching offerings in which students can qualify by opting for specific competences and a competence level at the Bachelor's or Master's level. The teaching offerings are pooled in two different catalogues for nontechnical complementary courses. The Learning Architecture consists of a cross-disciplinarily study offering. The centrally designed teaching offering ensures that courses in the nontechnical academic programms follow the specific profiling of TUHH degree courses. The learning architecture demands and trains independent educational planning as regards the individual development of competences. It also provides orientation knowledge in the form of "profiles". The subjects that can be studied in parallel throughout the student's entire study program - if need be, it can be studied in one to two semesters. In view of the adaptation problems that individuals commonly face in their first semesters after making the transition from school to university and in order to encourage individually planned semesters abroad, there is no obligation to study these subjects in one or two specific semesters during the course of studies. Teaching and Learning Arrangements provide for students, separated into B.Sc. and M.Sc., to learn with and from each other across semesters. The challenge of dealing with interdisciplinarity and a variety of stages of learning in courses are part of the learning architecture and are deliberately encouraged in specific courses. Fields of Teaching are based on research findings from the academic disciplines cultural studies, social studies, arts, historical studies, communication studies, migration studies and sustainability research, and from engineering didactics. In addition, from the winter semester 2014/15 students on all Bachelor's courses will have the opportunity to learn about business management and start-ups in a goal-oriented way. The fields of teaching are augmented by soft skills offers and a foreign language offer. Here, the focus is on encouraging goal-oriented communication skills, e.g. the skills required by outgoing engineers in international and intercultural situations. The Competence Level of the courses offered in this area is different as regards the basic training objective in the Bachelor's and Master's fields. These differences are reflected in the practical examples used, in content topics that refer to different professional application contexts, and in the higher scientific and theoretical level of abstraction in the B.Sc. This is also reflected in the different quality of soft skills, which relate to the different team positions and different group leadership functions of Bachelor's and Master's graduates in their future working life. Specialized Competence (Knowledge) Students can • explain specialized areas in context of the relevant non-technical disciplines, • outline basic theories, categories, terminology, models, concepts or artistic techniques in the disciplines represented in the learning area, • different specialist disciplines relate to their own discipline and differentiate it as well as make connections, • sketch the basic outlines of how scientific disciplines, paradigms, models, instruments, methods and forms of representation in the specialized sciences are subject to individual and socio-cultural interpretation and historicity, • Can communicate in a foreign language in a manner appropriate to the subject. <i>Skills</i> Professional Competence (Skills) In selected sub-areas students can • apply basic and specific methods of the said scientific disciplines, • question a specific technical phenomena, models, theories from the viewpoint of another, aforementioned specialist discipline, • to handle simple and advanced questions in aforementioned scientific disciplines in a successful manner, • justify their decisions on forms of organization and application in practical questions in contexts that go beyond the technical relationship to the subject.

Personal Competence <i>Social Competence</i> Personal Competences (Social Skills) Students will be able • to learn to collaborate in different manner, • to present and analyze problems in the abovementioned fields in a partner or group situation in a manner appropriate to the addressees, • to express themselves competently, in a culturally appropriate and gender-sensitive manner in the language of the country (as far as this study-focus would be chosen), • to explain nontechnical items to auditorium with technical background knowledge. <i>Autonomy</i> Personal Competences (Self-reliance) Students are able in selected areas • to reflect on their own profession and professionalism in the context of real-life fields of application • to organize themselves and their own learning processes • to reflect and decide questions in front of a broad education background • to communicate a nontechnical item in a competent way in written form or verbally • to organize themselves as an entrepreneurial subject country (as far as this study-focus would be chosen)	
Workload in Hours	Depends on choice of courses
Credit points	6

Courses
Information regarding lectures and courses can be found in the corresponding module handbook published separately.

Module M1311: Sustainable Water Management and Microbiology of Water Systems				
Courses				
Title	Typ		Hrs/wk	CP
Microbiology of water systems (L1782)	Lecture		2	3
Sustainable Water Management (L0406)	Project-/problem-based Learning		2	3
Module Responsible	Prof. Mathias Ernst			
Admission Requirements	None			
Recommended Previous Knowledge	Basic knowledge in water chemistry, Knowledge of main water treatment processes			
Educational Objectives	After taking part successfully, students have reached the following learning results			
Professional Competence	<i>Knowledge</i> Students will be able to explain the relevance of local and national water cycles on basis of water recycling targets. They will be able to separate into conventional and advanced treatment processes for both, drinking and wastewater treatment. Students are capable to name basic differences between water chemical parameters in drinking and wastewater analysis and define their significance for a sustainable water management. Students will be able to differentiate between natural and hygienically relevant bacteria in drinking water and will know modern microbiological methods for routine and scientific analyses of drinking water. They are familiar with the diverse microbiological processes in drinking water treatment and supply. The students know the legal regulations of the microbiological drinking water quality. <i>Skills</i> On basis of water use targets students will be able to prepare combinations of naturally based as well as technical water treatment processes. They will be able to calculate key parameters of treatment pathways for a water recycling study. Students will be able to deputise their conceptual design study by argumentation. Students will be capable to assess risks for the hygienic state of drinking water. Based on knowledge of methods they are able to evaluate results of routine analyses and research. Based on knowledge of processes, students will be able to suggest solutions to problems in drinking water supply. Personal Competence <i>Social Competence</i> Students will be able to work in diverse teams on problems in the field of sustainable water management. They will be able to coordinate complex tasks within their group and hand out duties accordingly. <i>Autonomy</i> Students will be in a position to work out presentations in the field of sustainable water management. They will be capable of finding creative solutions for water recycling concepts. Students will know how to use their technical knowledge for solving problems.			
Workload in Hours	Independent Study Time 124, Study Time in Lecture 56			
Credit points	6			
Course achievement	Compulsory	Bonus	Form	Description
	Yes	20 %	Presentation	
Examination	Written exam			
Examination duration and scale	90 min exam			
Assignment for the Following Curricula	Environmental Engineering: Core Qualification: Compulsory			

Course L1782: Microbiology of water systems	
Typ	Lecture
Hrs/wk	2
CP	3
Workload in Hours	Independent Study Time 62, Study Time in Lecture 28
Lecturer	Prof. Johannes Gescher, Prof. Mathias Ernst
Language	EN
Cycle	WiSe
Content	• Natural and hygienically relevant microorganisms in drinking water • Quantification of bacteria in drinking water • Identification of bacteria • Bacterial population analyses • Growth of bacteria and VBNC-state • Activity of bacteria in the environment • Biofilms in drinking water systems • Disinfection of drinking water and drinking water systems • Microbiological processes in drinking water treatment • Technical realization for optimized use of microbiological processes for drinking water production • Impact factors on microbiological drinking water quality during distribution and compliance with legal requirements on hygiene at the consumer's tap
Literature	• Allgemeine Mikrobiologie. 2007. Fuchs, G. (Hrsg.), 8. Aufl., Thieme Verlag, Stuttgart. • Brock Biology of Microorganisms. 2015. Madigan, M. T., Martinko, J. M., Bender, K. S., Buckley, D. H., and Stahl, D. A. (eds.), 14. edition, Pearson Education Ltd, Harlow, UK. • Microbial growth in drinking- water supplies: Problems, causes control and research needs. 2014. Van der Kooij, D. and Van der Wielen, P. W. J. J. (eds.) IWA Publishing, London.

Course L0406: Sustainable Water Management	
Typ	Project-/problem-based Learning
Hrs/wk	2
CP	3
Workload in Hours	Independent Study Time 62, Study Time in Lecture 28
Lecturer	Prof. Mathias Ernst
Language	EN
Cycle	WiSe
Content	The course provides knowledge on the sustainable treatment and management of the resource water. Used water is an alternative resource and can be recycled in any field of the urban water cycle after adequate treatment. The resulting water quality is the decisive issue. In the course the central quality parameters of drinking- as well as wastewater assessment will be presented and discussed. Moreover the legal frame for water reuse in the EU and examples from all over the world will be communicated. The students receive the task to develop a conceptual design study of an indirect potable reuse facility in given boundary conditions. To fulfill this task, the students will work in small groups representing a consulting firm. Later in the course the firms will present their concepts. In preparation to the team presentation further knowledge on alternative water resources and sustainable management will be provided. International case studies will be presented and discussed. Next to the communication of technical details, planning tools for the implementation of alternative water management will be given also Option for an effective public perception program of later water users.
Literature	• Milestones in Water Reuse, V. Lazarova, T. Asano, A. Bahri, J. Anderson, IWA Publishing 2013 • Current UN World Water Development Reports • Water Security for Better Lives, OECD Studie 2013 • PPT's provided during the course

Module M1313: Fluid Mechanics, Hydraulics and Geo-Information-Systems in Water Management				
Courses				
Title		Typ	Hrs/wk	CP
Geo-Information-Systems in Water Management and Hydraulic Engineering (L0963)		Project-/problem-based Learning	2	2
Fluid Mechanics and Hydraulics (L1246)		Lecture	2	2
Fluid Mechanics and Hydraulics (L1656)		Recitation Section (small)	1	2
Module Responsible	Prof. Peter Fröhle			
Admission Requirements	None			
Recommended Previous Knowledge	Mathematics (calculus) and physics; Knowledge of statics and thermodynamik would be beneficial.			
Educational Objectives	After taking part successfully, students have reached the following learning results			
Professional Competence	<i>Knowledge</i> After finishing the module the students will learn the properties of fluid, hydrostatics, Fluid kinematics, conservation equations (mass, energy and momentum), flow in pipes, boundary layer theory, viscous flow (skin friction and drag forces), flow in pipes, hydraulics of open channel, flow in compound and natural channels, energy head losses. <i>Skills</i> The students will be capable to calculate and analyse the forces in the fluids as well as flow in pipes and channels. Personal Competence <i>Social Competence</i> The students learn to deploy their knowledge in applied problems such as calculation of water level and the rate of water rise in flood events. Furthermore, they will be able to work in team with engineers of other disciplines, for instance by designing of gates. <i>Autonomy</i> The students will be able to independently extend their knowledge and apply it to new problems.			
Workload in Hours	Independent Study Time 110, Study Time in Lecture 70			
Credit points	6			
Course achievement	None			
Examination	Written exam			
Examination duration and scale	150 minutes including definition and descriptions as well as calculations			
Assignment for the Following Curricula	Environmental Engineering: Core Qualification: Compulsory			

Course L0963: Geo-Information-Systems in Water Management and Hydraulic Engineering	
Typ	Project-/problem-based Learning
Hrs/wk	2
CP	2
Workload in Hours	Independent Study Time 32, Study Time in Lecture 28
Lecturer	Prof. Peter Fröhle
Language	EN
Cycle	WiSe
Content	Theoretical basics of Geo-Information-Systems • Data models, geographical coordinates, geo-referencing, map-views • Data mining and - analyses of geo-data • Analysis techniques
Literature	None

Course L1246: Fluid Mechanics and Hydraulics	
Typ	Lecture
Hrs/wk	2
CP	2
Workload in Hours	Independent Study Time 32, Study Time in Lecture 28
Lecturer	Dr. Mohammad Hassan Naser Moaddeli
Language	EN
Cycle	WiSe
Content	Properties of fluid, hydrostatics, Fluid kinematics, conservation equations (mass, energy and momentum), flow in pipes, boundary layer theory of laminar and turbulent flow, viscous flow (skin friction and drag forces), open channel hydraulics, flow in compound and natural channels, local energy head losses
Literature	R.L. Street, G.Z. Watters, J.K. Vennard: Elementary Fluid Mechanics, 7th edition, 1996 Chow, V.T., Open Channel hydraulics, Van Te Chow, 1988

Course L1656: Fluid Mechanics and Hydraulics	
Typ	Recitation Section (small)
Hrs/wk	1
CP	2
Workload in Hours	Independent Study Time 46, Study Time in Lecture 14
Lecturer	Dr. Mohammad Hassan Nasermoaddeli
Language	EN
Cycle	WiSe
Content	See interlocking course
Literature	See interlocking course

Module M1312: Environmental Analysis and Water Technology Practice				
Courses				
Title		Typ	Hrs/wk	CP
Practical Course in Water and Wastewater Technology I (L0503)		Practical Course	3	3
Environmental Analysis (L0354)		Lecture	2	3
Module Responsible	Dr. Dorothea Rechtenbach			
Admission Requirements	None			
Recommended Previous Knowledge	Basic knowledge in chemistry and physics (knowledge required at school)			
Educational Objectives	After taking part successfully, students have reached the following learning results			
Professional Competence	The students know basic analytical procedures for evaluating the quality of different environmental compartments. The students are able to understand and to practically apply methodologies for environmental analysis as well as descriptions of experiments and experimental setups in wasterwater analysis. The students are able to organize working processes within a team in a targeted way and based on the divison of labour. The students are able to independently exploit sources and conduct experiments following written procedures without external assistance.			
<i>Knowledge</i>				
<i>Skills</i>				
Personal Competence				
<i>Social Competence</i>				
<i>Autonomy</i>				
Workload in Hours	Independent Study Time 110, Study Time in Lecture 70			
Credit points	6			
Course achievement	None			
Examination	Written exam			
Examination duration and scale	90 minutes written exam including written report for the practical			
Assignment for the Following Curricula	Environmental Engineering: Core Qualification: Compulsory			

Course L0503: Practical Course in Water and Wastewater Technology I	
Typ	Practical Course
Hrs/wk	3
CP	3
Workload in Hours	Independent Study Time 48, Study Time in Lecture 42
Lecturer	Dr. Dorothea Rechtenbach
Language	EN
Cycle	WiSe
Content	- Impact of pretreatment of wastewater samples on analytical results - Analysis of nutrients in wastewater samples (different methods for nitrate analysis) - Alkalinity - TOC, COD - microscopic analysis of microorganisms relevant in wastewater treatment
Literature	Skript auf StudIP

Course L0354: Environmental Analysis	
Typ	Lecture
Hrs/wk	2
CP	3
Workload in Hours	Independent Study Time 62, Study Time in Lecture 28
Lecturer	Dr. Dorothea Rechtenbach, Dr. Henning Mangels
Language	EN
Cycle	WiSe
Content	Introduction Sampling in different environmental compartments, sample transportation, sample storage Sample preparation Photometry Wastewater analysis Introduction into chromatography Gas chromatography HPLC Mass spectrometry Optical emission spectrometry Atom absorption spectrometry Quality assurance in environmental analysis
Literature	Roger Reeve, Introduction to Environmental Analysis, John Wiley & Sons Ltd., 2002 (TUB: USD-728) Pradyot Patnaik, Handbook of environmental analysis: chemical pollutants in air, water, soil, and solid wastes, CRC Press, Boca Raton, 2010 (TUB: USD-716) Chunlong Zhang, Fundamentals of Environmental Sampling and Analysis, John Wiley & Sons Ltd., Hoboken, New Jersey, 2007 (TUB: USD-741) Miroslav Radojević, Vladimir N. Bashkin, Practical Environmental Analysis RSC Publ., Cambridge, 2006 (TUB: USD-720) Werner Funk, Vera Dammann, Gerhild Donnevert, Sarah Iannelli (Translator), Eric Iannelli (Translator), Quality Assurance in Analytical Chemistry: Applications in Environmental, Food and Materials Analysis, Biotechnology, and Medical Engineering, 2nd Edition, WILEY-VCH Verlag GmbH & Co. KGaA, Weinheim, 2007 (TUB: CHF-350) STANDARD METHODS FOR THE EXAMINATION OF WATER AND WASTEWATER, 21st Edition, Andrew D. Eaton, Leonore S. Clesceri, Eugene W. Rice, and Arnold E. Greenberg, editors, 2005 (TUB: CHF-428) K. Robards, P. R. Haddad, P. E. Jackson, Principles and Practice of Modern Chromatographic Methods, Academic Press G. Schwedt, Chromatographische Trennmethoden, Thieme Verlag H. M. McNair, J. M. Miller, Basic Gas Chromatography, Wiley W. Gottwald, GC für Anwender, VCH B. A. Bidlingmeyer, Practical HPLC Methodology and Applications, Wiley K. K. Unger, Handbuch der HPLC, GIT Verlag G. Aced, H. J. Möckel, Liquidchromatographie, VCH Charles B. Boss and Kenneth J. Fredeen, Concepts, Instrumentation and Techniques in Inductively Coupled Plasma Optical Emission Spectrometry Perkin-Elmer Corporation 1997, On-line available at: http://files.instrument.com.cn/bbs/upfile/2006291448.pdf Atomic absorption spectrometry: theory, design and applications, ed. by S. J. Haswell 1991 (TUB: 2727-5614) Royal Society of Chemistry, Atomic absorption spectrometry (http://www.kau.edu.sa/Files/130002/Files/6785_AAs.pdf)

Module M2003: Biological Waste Treatment				
Courses				
Title	Typ		Hrs/wk	CP
Waste and Environmental Chemistry (L0328)	Practical Course		2	2
Biological Waste Treatment (L0318)	Project-/problem-based Learning		3	4
Module Responsible	Prof. Kerstin Kuchta			
Admission Requirements	None			
Recommended Previous Knowledge	chemical and biological basics			
Educational Objectives	After taking part successfully, students have reached the following learning results			
Professional Competence				
<i>Knowledge</i>	The module aims possess knowledge concerning the planning of biological waste treatment plants. Students are able to explain the design and layout of anaerobic and aerobic waste treatment plants in detail, describe different techniques for waste gas treatment plants for biological waste treatment plants and explain different methods for waste analytics.			
<i>Skills</i>	The students are able to discuss the compilation of design and layout of plants. They can critically evaluate techniques and quality control measurements. The students can recherche and evaluate literature and data connected to the tasks given in der module and plan additional tests. They are capable of reflecting and evaluating findings in the group.			
Personal Competence				
<i>Social Competence</i>	Students can participate in subject-specific and interdisciplinary discussions, develop cooperated solutions and defend their own work results in front of others and promote the scientific development in front of colleagues. Furthermore, they can give and accept professional constructive criticism.			
<i>Autonomy</i>	Students can independently tap knowledge from literature, business or test reports and transform it to the course projects. They are capable, in consultation with supervisors as well as in the interim presentation, to assess their learning level and define further steps on this basis. Furthermore, they can define targets for new application-or research-oriented duties in accordance with the potential social, economic and cultural impact.			
Workload in Hours	Independent Study Time 110, Study Time in Lecture 70			
Credit points	6			
Course achievement	Compulsory	Bonus	Form	Description
	Yes	None	Subject	theoretical and practical work
Examination	Presentation			
Examination duration and scale	Elaboration and Presentation (15-25 minutes in groups)			
Assignment for the Following Curricula	Civil Engineering: Specialisation Coastal Engineering: Elective Compulsory Civil Engineering: Specialisation Geotechnical Engineering: Elective Compulsory Civil Engineering: Specialisation Structural Engineering: Elective Compulsory Civil Engineering: Specialisation Water and Traffic: Elective Compulsory Bioprocess Engineering: Specialisation A - General Bioprocess Engineering: Elective Compulsory Chemical and Bioprocess Engineering: Specialisation General Process Engineering: Elective Compulsory Chemical and Bioprocess Engineering: Specialisation Chemical Process Engineering: Elective Compulsory Chemical and Bioprocess Engineering: Specialisation Chemical and Bioprocess Engineering: Elective Compulsory Chemical and Bioprocess Engineering: Specialisation Bioprocess Engineering: Elective Compulsory Chemical and Bioprocess Engineering: Specialisation Chemical and Bioprocess Engineering: Elective Compulsory Environmental Engineering: Core Qualification: Compulsory International Management and Engineering: Specialisation II. Renewable Energy: Elective Compulsory Process Engineering: Specialisation Environmental Process Engineering: Elective Compulsory Water and Environmental Engineering: Specialisation Cities: Elective Compulsory Water and Environmental Engineering: Specialisation Environment: Elective Compulsory			

Course L0328: Waste and Environmental Chemistry	
Typ	Practical Course
Hrs/wk	2
CP	2
Workload in Hours	Independent Study Time 32, Study Time in Lecture 28
Lecturer	Prof. Kerstin Kuchta
Language	EN
Cycle	WiSe
Content	The participants are divided into groups. Each group prepares a transcript on the experiment performed, which is then used as basis for discussing the results and to evaluate the performance of the group and the individual student. In some experiments the test procedure and the results are presented in seminar form, accompanied by discussion and results evaluation. Experiments are e.g. Screening and particle size determination Fos/Tac AAS Calorific value
Literature	Scripte

Course L0318: Biological Waste Treatment	
Typ	Project-/problem-based Learning
Hrs/wk	3
CP	4
Workload in Hours	Independent Study Time 78, Study Time in Lecture 42
Lecturer	Prof. Kerstin Kuchta
Language	EN
Cycle	WiSe
Content	1. Introduction 2. biological basics 3. determination process specific material characterization 4. aerobic degradation (Composting, stabilization) 5. anaerobic degradation (Biogas production, fermentation) 6. Technical layout and process design 7. Flue gas treatment 8. Plant design practical phase
Literature	

Module M2033: Subsurface Processes			
Courses			
Title	Typ	Hrs/wk	CP
Modeling of Subsurface Processes (L2731)	Recitation Section (small)	3	3
Subsurface Solute Transport (L2728)	Lecture	2	2
Subsurface Solute Transport (L2729)	Recitation Section (large)	1	1
Module Responsible	Dr. Milad Aminzadeh		
Admission Requirements	None		
Recommended Previous Knowledge	Basic Mathematics, Hydrology		
Educational Objectives	After taking part successfully, students have reached the following learning results		
Professional Competence	Upon completion of this module, the students will understand the mechanisms controlling solute transport in soil and natural porous media and will be able to work with the equations that govern the fate and transport of solutes in porous media. Analytical, numerical and experimental tools and techniques will be used in this module. In addition to the physical insights, the students will be exposed to analytical, experimental and numerical tools and techniques in this module. This provides them with an excellent opportunity to improve their skills on multiple fronts which will be useful in their future career. Teamwork & problem solving The students will be involved in writing individual reports and presentation. This will contribute to the students' ability and willingness to work independently and responsibly.		
<i>Knowledge</i>			
<i>Skills</i>			
Personal Competence			
<i>Social Competence</i>			
<i>Autonomy</i>			
Workload in Hours	Independent Study Time 96, Study Time in Lecture 84		
Credit points	6		
Course achievement	None		
Examination	Subject theoretical and practical work		
Examination duration and scale	Report		
Assignment for the Following Curricula	Civil Engineering: Specialisation Structural Engineering: Elective Compulsory Civil Engineering: Specialisation Geotechnical Engineering: Elective Compulsory Civil Engineering: Specialisation Coastal Engineering: Elective Compulsory Civil Engineering: Specialisation Water and Traffic: Elective Compulsory Civil Engineering: Specialisation Computational Engineering: Elective Compulsory Chemical and Bioprocess Engineering: Technical Complementary Course: Elective Compulsory Chemical and Bioprocess Engineering: Technical Complementary Course: Elective Compulsory Environmental Engineering: Core Qualification: Compulsory Process Engineering: Specialisation Environmental Process Engineering: Elective Compulsory Process Engineering: Specialisation Process Engineering: Elective Compulsory Water and Environmental Engineering: Specialisation Water: Compulsory Water and Environmental Engineering: Specialisation Environment: Elective Compulsory		

Course L2731: Modeling of Subsurface Processes	
Typ	Recitation Section (small)
Hrs/wk	3
CP	3
Workload in Hours	Independent Study Time 48, Study Time in Lecture 42
Lecturer	Mohammad Aziz Zarif
Language	EN
Cycle	WiSe
Content	Basic usage and background of chosen computer software to calculate flow and transport in the saturated and unsaturated zone and to analyze field data like pumping test data
Literature	siehe korrespondierende Vorlesung

Course L2728: Subsurface Solute Transport	
Typ	Lecture
Hrs/wk	2
CP	2
Workload in Hours	Independent Study Time 32, Study Time in Lecture 28
Lecturer	Dr. Milad Aminzadeh
Language	EN
Cycle	WiSe
Content	Basic physical properties of soil: Definition and quantification; Liquid flow in soils (Darcy's law); Solute transport in soils; Practical analysis to measure dispersion coefficient in soil under different boundary conditions; Advanced topics (e.g. Application of Artificial Intelligence to predict soil salinization)
Literature	- Environmental Soil Physics, by Daniel Hillel - Soil Physics, Sixth Edition, by William A. Jury and Robert Horton

Course L2729: Subsurface Solute Transport	
Typ	Recitation Section (large)
Hrs/wk	1
CP	1
Workload in Hours	Independent Study Time 16, Study Time in Lecture 14
Lecturer	Dr. Milad Aminzadeh
Language	EN
Cycle	WiSe
Content	See interlocking course
Literature	See interlocking course

Module M0828: Urban Environmental Management				
Courses				
Title		Type	Hrs/wk	CP
Noise Protection (L1109)		Lecture	2	2
Urban Infrastructures (L0874)		Project-/problem-based Learning	3	4
Module Responsible	Dr. Dorothea Rechtenbach			
Admission Requirements	None			
Recommended Previous Knowledge	• Knowledge on Urban planning • Knowledge on measures for climate protection • General knowledge of scientific writing/working			
Educational Objectives	After taking part successfully, students have reached the following learning results			
Professional Competence	<i>Knowledge</i> Students can describe urban development corridors as well as current and future urban environmental problems. They are able to explain the causes of environmental problems (like noise). Students can specify applications for various technical innovations and explain why these contribute to the improvement of urban life. They can, for example, derive and discuss measures for effective noise abatement. <i>Skills</i> Students are able to develop specific solutions for correcting existing or future environment-related problems of urban development. They can define a range of conceptual and technical solutions for environmental problems for different development paths. To solve specific urban environmental problems they can select technical innovations and integrate them into the urban context. Personal Competence <i>Social Competence</i> The students can work together in international groups. <i>Autonomy</i> Students are able to organize their work flow to prepare themselves for presentations and contributions to the discussions. They can acquire appropriate knowledge by making enquiries independently.			
Workload in Hours	Independent Study Time 110, Study Time in Lecture 70			
Credit points	6			
Course achievement	None			
Examination	Written elaboration			
Examination duration and scale	Written Report plus oral Presentation			
Assignment for the Following Curricula	Civil Engineering: Specialisation Structural Engineering: Elective Compulsory Civil Engineering: Specialisation Geotechnical Engineering: Elective Compulsory Civil Engineering: Specialisation Coastal Engineering: Elective Compulsory Civil Engineering: Specialisation Water and Traffic: Elective Compulsory Environmental Engineering: Core Qualification: Elective Compulsory Joint European Master in Environmental Studies - Cities and Sustainability: Core Qualification: Compulsory Logistics, Infrastructure and Mobility: Specialisation Infrastructure and Mobility: Elective Compulsory Water and Environmental Engineering: Specialisation Environment: Elective Compulsory Water and Environmental Engineering: Specialisation Cities: Compulsory			

Course L1109: Noise Protection	
Type	Lecture
Hrs/wk	2
CP	2
Workload in Hours	Independent Study Time 32, Study Time in Lecture 28
Lecturer	Prof. Martin Jäschke
Language	EN
Cycle	SoSe
Content	
Literature	1) Müller & Möser (2013): Handbook of Engineering Acoustics (also available in German) 2) WHO (1999): Guidelines for Community Noise 3) Environmental Noise Directive 2002/49/EG 4) ISO 9613-2 (1996): Acoustics, Attenuation of sound during propagation outdoors, Part 2: General method of calculation

Course L0874: Urban Infrastructures	
Typ	Project-/problem-based Learning
Hrs/wk	3
CP	4
Workload in Hours	Independent Study Time 78, Study Time in Lecture 42
Lecturer	Dr. Dorothea Rechtenbach
Language	EN
Cycle	SoSe
Content	Problem Based Learning Main topics are: • Central vs. Decentral Wastewater Treatment. • Compaction of Cities. • Car Free Cities. • Multifunctional Places in Cities. • The Sustainability of Freight Transport in Cities.
Literature	Depends on chosen topic.

Module M0870: Management of Surface Water				
Courses				
Title	Typ		Hrs/wk	CP
Modelling of Flow in Rivers and Estuaries (L0810)	Lecture		3	4
Nature-Oriented Hydraulic Engineering / Integrated Flood Protection (L0961)	Project-/problem-based Learning		2	2
Module Responsible	Prof. Peter Fröhle			
Admission Requirements	None			
Recommended Previous Knowledge	Fundamentals of Hydromechanics, Hydraulics, Hydrology and Hydraulic Engineering; Hydraulic Engineering I and Hydraulic Engineering II			
Educational Objectives	After taking part successfully, students have reached the following learning results			
Professional Competence	<i>Knowledge</i> Students are able to define in detail the basic processes that are related to the modelling of flows in hydraulic engineering. Besides, they can describe the basic aspects of numerical modelling and actual numerical models for the simulation of flows and waves. They can also depict the concepts of nature oriented hydraulic engineering. <i>Skills</i> Students are able to apply hydrodynamic-numerical models to practical hydraulic engineering tasks. Furthermore, the students are able to set up flood-risk management concepts and are able to apply basic concepts of renaturation to practical problems. Personal Competence <i>Social Competence</i> The students are able to deploy their gained knowledge in applied problems of the practical nature-based hydraulic engineering. Additionally, they will be able to work in team with engineers of other disciplines. <i>Autonomy</i> The students will be able to independently extend their knowledge and apply it to new problems.			
Workload in Hours	Independent Study Time 110, Study Time in Lecture 70			
Credit points	6			
Course achievement	None			
Examination	Written exam			
Examination duration and scale	The duration of the examination is 150 min. The examination includes tasks with respect to the general understanding of the lecture contents and calculations tasks.			
Assignment for the Following Curricula	Civil Engineering: Specialisation Water and Traffic: Compulsory Environmental Engineering: Core Qualification: Elective Compulsory Joint European Master in Environmental Studies - Cities and Sustainability: Core Qualification: Compulsory Water and Environmental Engineering: Specialisation Water: Compulsory Water and Environmental Engineering: Specialisation Environment: Compulsory Water and Environmental Engineering: Specialisation Cities: Elective Compulsory			

Course L0810: Modelling of Flow in Rivers and Estuaries	
Typ	Lecture
Hrs/wk	3
CP	4
Workload in Hours	Independent Study Time 78, Study Time in Lecture 42
Lecturer	Prof. Edgar Nehlsen, Prof. Peter Fröhle
Language	EN
Cycle	SoSe
Content	Introduction to numerical flow modelling - Processes affecting the flow - Examples and applications of numerical models - Procedure of numerical modelling - Model concept Basic equations of hydrodynamics - Saint-Venant equations - Euler Equations - Navier-Stokes equations - Reynolds-averaged Navier-Stokes equations - Shallow water equations Solving schemes - Numerical discretization - Solution algorithms - Convergence
Literature	Vorlesungsskript Literaturempfehlungen Bund der Ingenieure für Wasserwirtschaft, Abfallwirtschaft und Kulturbau (1997): Hydraulische Berechnung von naturnahen Fließgewässern. Düsseldorf: BWK (BWK-Merkblatt). Chow, Ven-te (1959): Open-channel Hydraulics. New York usw.: McGraw-Hill (McGraw-Hill Civil Engineering Series). Deutsche Vereinigung für Wasserwirtschaft, Abwasser und Abfall e.V. (DWA); DWA-Arbeitsgruppe WW-3.2 Mehrdimensionale numerische Modelle, DWA-Arbeitsgruppe WW-3.2 Mehrdimensionale numerische (2019a): Merkblatt DWA-M 543-2 Geodaten in der Fließgewässermodellierung Teil 1: Geodaten in der Fließgewässermodellierung. Februar 2019. Hennef: Deutsche Vereinigung für Wasserwirtschaft, Abwasser und Abfall (DWA-Regelwerk, 543-1). Deutsche Vereinigung für Wasserwirtschaft, Abwasser und Abfall e.V. (DWA); DWA-Arbeitsgruppe WW-3.2 Mehrdimensionale numerische Modelle, DWA-Arbeitsgruppe WW-3.2 Mehrdimensionale numerische (2019b): Merkblatt DWA-M 543-2 Geodaten in der Fließgewässermodellierung Teil 2: Bedarfsgerechte Datenerfassung und -aufbereitung. Februar 2019. Hennef: Deutsche Vereinigung für Wasserwirtschaft, Abwasser und Abfall (DWA-Regelwerk, 543-2). Deutsche Vereinigung für Wasserwirtschaft, Abwasser und Abfall e.V. (DWA); DWA-Arbeitsgruppe WW-3.2 Mehrdimensionale numerische Modelle, DWA-Arbeitsgruppe WW-3.2 Mehrdimensionale numerische (2019c): Merkblatt DWA-M 543-3 Geodaten in der Fließgewässermodellierung - Teil 3: Aspekte der Strömungsmodellierung und Fallbeispiele. Februar 2019. Hennef: Deutsche Vereinigung für Wasserwirtschaft, Abwasser und Abfall (DWA-Regelwerk, 543-3). Hervouet, Jean-Michel (2007): Hydrodynamics of free surface flows. Modelling with the finite element method. Chichester: Wiley. Online verfügbar unter http://www.loc.gov/catdir/enhancements/fy0741/2007296953-b.html. IAHR (2015): Professional Specifications for Physical and Numerical Studies in Environmental Hydraulics. In: Hydrolink (3/2015), S. 90-92. Olsen, Nils Reidar B. (2012): Numerical Modelling and Hydraulics. 3. Aufl. Department of Hydraulic and Environmental Engineering, The Norwegian University of Science and Technology. Szymkiewicz, Romuald (2010): Numerical modeling in open channel hydraulics. Dordrecht: Springer (Water science and technology library, 83). van Waveren, Harold (1999-): Good modelling practice handbook. [Utrecht], Lelystad, Den Haag: STOWA; Rijkswaterstaat-RIZA; SDU, afd. SEO/RIZA [etc. distr.] (Nota, nr. 99.036). Zielke, Werner (Hg.) (1999): Numerische Modelle von Flüssen, Seen und Küstengewässern. Deutscher Verband für Wasserwirtschaft und Kulturbau. Bonn: Wirtschafts- und Verl.-Ges. Gas und Wasser (Schriftenreihe des Deutschen Verbandes für Wasserwirtschaft und Kulturbau, 127).

Course L0961: Nature-Oriented Hydraulic Engineering / Integrated Flood Protection	
Typ	Project-/problem-based Learning
Hrs/wk	2
CP	2
Workload in Hours	Independent Study Time 32, Study Time in Lecture 28
Lecturer	Dr. Natasa Manojlovic, Prof. Peter Fröhle
Language	EN
Cycle	SoSe
Content	• Regime-Theory and application for the development of environmental guiding principles of rivers • Engineering - biological measures for the stabilization of rivers • Risk management in flood protection • Design techniques in technical flood protection • Methods for the assessment of flood caused damages
Literature	Vorlesungsumdruck

Module M0871: Hydrological Systems				
Courses				
Title	Typ	Hrs/wk	CP	
Applied Surface Hydrology (L0289)	Lecture	2	2	
Applied Surface Hydrology (L1412)	Project-/problem-based Learning	1	2	
Interaction Water - Environment in Fluvial Areas (L0295)	Project-/problem-based Learning	1	2	
Module Responsible	Prof. Peter Fröhle			
Admission Requirements	None			
Recommended Previous Knowledge	Fundamentals of Hydromechanics and Hydraulic Engineering: Hydraulic Engineering I and Hydraulic Engineering II			
Educational Objectives	After taking part successfully, students have reached the following learning results			
Professional Competence	<i>Knowledge</i> The students are able to define the basic concepts of hydrology and water management. They are able to describe and quantify the relevant processes of the hydrological water cycle. Besides, the students know the main aspects of rainfall-run-off-models and are able to theoretically derive established reservoir / storage models and a unit-hydrograph. <i>Skills</i> The students are able to use the basic hydrological concepts and approaches and are able to theoretically derive established reservoir / storage models or a unit-hydrograph as the basis for rainfall-run-off-models. The student are able to explain the basic concepts of measurements of hydrological and hydrodynamic values in nature and are able to perform, analyze and statistically assess these measurements. Furthermore, they are able to apply a hydrological model to basic hydrological problems. Personal Competence <i>Social Competence</i> The students are able to deploy their gained knowledge in applied problems of the hydrology and water management. Additionally, they will be able to work in team with engineers of other disciplines. <i>Autonomy</i> The students will be able to independently extend their knowledge and apply it to new problems			
Workload in Hours	Independent Study Time 124, Study Time in Lecture 56			
Credit points	6			
Course achievement	None			
Examination	Written exam			
Examination duration and scale	The duration of the examination is 90 min. The examination includes tasks with respect to the general understanding of the lecture contents and calculations tasks.			
Assignment for the Following Curricula	Civil Engineering: Specialisation Computational Engineering: Elective Compulsory Civil Engineering: Specialisation Water and Traffic: Compulsory Environmental Engineering: Core Qualification: Elective Compulsory Joint European Master in Environmental Studies - Cities and Sustainability: Core Qualification: Compulsory Water and Environmental Engineering: Specialisation Cities: Elective Compulsory Water and Environmental Engineering: Specialisation Environment: Elective Compulsory Water and Environmental Engineering: Specialisation Water: Elective Compulsory			

Course L0289: Applied Surface Hydrology	
Typ	Lecture
Hrs/wk	2
CP	2
Workload in Hours	Independent Study Time 32, Study Time in Lecture 28
Lecturer	Prof. Peter Fröhle
Language	EN
Cycle	SoSe
Content	Basics of hydrology: - Hydrological cycle - Data acquisition - Data analyses and statistical assessment - Statistics of extremes - Regionalization methods for hydrological values - Rainfall-run-off modelling on the basis of a unit hydrograph concepts - Application of rainfall-run-off models on the basis of Kalypso-Hydrology which is an OpenSource Software Tool.
Literature	http://de.wikipedia.org/wiki/Kalypso_(Software) http://kalypso.bjoernsen.de/ http://sourceforge.net/projects/kalypso/

Course L1412: Applied Surface Hydrology	
Typ	Project-/problem-based Learning
Hrs/wk	1
CP	2
Workload in Hours	Independent Study Time 46, Study Time in Lecture 14
Lecturer	Prof. Peter Fröhle
Language	EN
Cycle	SoSe
Content	See interlocking course
Literature	See interlocking course

Course L0295: Interaction Water - Environment in Fluvial Areas	
Typ	Project-/problem-based Learning
Hrs/wk	1
CP	2
Workload in Hours	Independent Study Time 46, Study Time in Lecture 14
Lecturer	Prof. Peter Fröhle
Language	EN
Cycle	SoSe
Content	A problem based learning course. The problem will be solved by the students more or less self-contained. The topics will be introduced and elaborated over the semester.
Literature	-

Module M1123: Selected Topics in Environmental Engineering			
Courses			
Title	Typ	Hrs/wk	CP
Environmental Aquatic Chemistry (L1444)	Lecture	2	3
Solid Matter Process Technology for Biomass (L0052)	Lecture	2	3
Sustainable landfill design and operation (L3270)	Integrated Lecture	3	3
Sludge Treatment (L0520)	Lecture	2	3
Special topics of the Environmental engineering 1CP (L3289)		1	1
Special topics of the Environmental engineering 2CP (L3290)		2	2
Special topics of the Environmental engineering 3CP (L3291)		3	3
Thermal Biomass Utilization (L1767)	Lecture	2	2
Thermal Biomass Utilization (L2386)	Practical Course	1	1
Module Responsible	Prof. Mathias Ernst		
Admission Requirements	None		
Recommended Previous Knowledge			
Educational Objectives	After taking part successfully, students have reached the following learning results		
Professional Competence <i>Knowledge</i> <i>Skills</i> Personal Competence <i>Social Competence</i> <i>Autonomy</i>			
Workload in Hours	Depends on choice of courses		
Credit points	6		
Assignment for the Following Curricula	Environmental Engineering: Core Qualification: Elective Compulsory Water and Environmental Engineering: Specialisation Cities: Elective Compulsory Water and Environmental Engineering: Specialisation Environment: Elective Compulsory Water and Environmental Engineering: Specialisation Water: Elective Compulsory		

Course L1444: Environmental Aquatic Chemistry	
Typ	Lecture
Hrs/wk	2
CP	3
Workload in Hours	Independent Study Time 62, Study Time in Lecture 28
Examination Form	Klausur
Examination duration and scale	60 min
Lecturer	Dr. Klaus Johannsen
Language	EN
Cycle	SoSe
Content	• Concentration and activity • Gas-water partitioning • Acid/base equilibria • Alkalinity and acidity • Precipitation/dissolution equilibria • Redox equilibria • Complex formation • Sorption
Literature	Worch, E.: Hydrochemistry. Basic Concepts and Exercises. De Gruyter, Berlin, 2015

Course L0052: Solid Matter Process Technology for Biomass	
Typ	Lecture
Hrs/wk	2
CP	3
Workload in Hours	Independent Study Time 62, Study Time in Lecture 28
Examination Form	Klausur
Examination duration and scale	60 min
Lecturer	Prof. Werner Sitzmann
Language	DE
Cycle	SoSe
Content	The industrial application of unit operations as part of process engineering is explained by actual examples of solid biomass processes. Size reduction, transportation and dosing, drying and agglomeration of renewable resources are described as important unit operations when producing solid fuels and bioethanol, producing and refining edible oils, when making Btl - and WPC - products. Aspects of explosion protection and plant design complete the lecture.
Literature	Kaltschmitt M., Hartmann H. (Hrsg.): Energie aus Bioamasse, Springer Verlag, 2001, ISBN 3-540-64853-4 Bundesministerium für Ernährung, Landwirtschaft und Verbraucherschutz, Schriftenreihe Nachwachsende Rohstoffe, Fachagentur Nachwachsende Rohstoffe e.V. www.nachwachsende-rohstoffe.de Bockisch M.: Nahrungsfette und -öle, Ulmer Verlag, 1993, ISBN 380000158175

Course L3270: Sustainable landfill design and operation	
Typ	Integrated Lecture
Hrs/wk	3
CP	3
Workload in Hours	Independent Study Time 48, Study Time in Lecture 42
Examination Form	Klausur
Examination duration and scale	60 min
Lecturer	Dr. Marco Ritzkowski
Language	EN
Cycle	SoSe
Content	The course introduces the development of modern waste resource management and demonstrates the importance of landfills in the context of recycling processes. Based on international (EU) and national legislation, the current landfill situation is presented and the future significance of landfills will be discussed. A central element of the course deals with the main transformation processes in the landfilled waste, the emission of gases and leachate, the long-term behaviour of landfills as well as aftercare and after-utilisation measures. Further focal points of the course are measures for the sustainable reduction of environmentally and climate-damaging emissions and aspects of landfill technology in an international context.
Literature	1) Waste Management. Bernd Bilitewski; Georg Härdtle; Klaus Marek (Eds.), ISBN: 9783540592105 , Springer Verlag Lehrbuchsammlung der TUB, Signatur USH-305 2) Solid Waste Technology and Management. Thomas Christensen (Ed.), ISBN: 978-1-4051-7517-3 , Wiley Verlag Lesesaal 2: US - Umweltschutz, Signatur USH-332 3) Solid Waste Landfilling - Concepts, Processes, Technologies. Cossu, R. and Stegmann, R. (Eds.), ISBN: 978-0-12-818336-6 PDF (Volltext) über TUB

Course L0520: Sludge Treatment	
Typ	Lecture
Hrs/wk	2
CP	3
Workload in Hours	Independent Study Time 62, Study Time in Lecture 28
Examination Form	Klausur
Examination duration and scale	60 min
Lecturer	Dr. Joachim Behrendt
Language	EN
Cycle	SoSe
Content	Sedimentation characteristic and thickening, Centrifugation, Flotation, Filtration, Aerobic sludge stabilisation, Sludge Digestion, Sludge Disintegration, Sludge Dewatering, Natural Processes for Sludge Treatment, Nutrient Recovery from Sludge, Thermal Processes and Incineration.
Literature	Tchobanoglous, George (Metcalf & Eddy, Inc., ;) Wastewater engineering : treatment and reuse ISBN: 0070418780 (alk. paper) ISBN: 0071122508 (ISE (*pbk)) Boston [u.a.] : McGraw-Hill, 2003 TUB_HH_Katalog Cleverson Vitorio Andreoli, Marcos von Sperling, Fernando Fernandes Sludge Treatment and Disposal ISBN 9781843391661 IWA Publishing, 2007

Course L3289: Special topics of the Environmental engineering 1CP	
Typ	
Hrs/wk	1
CP	1
Workload in Hours	Independent Study Time 16, Study Time in Lecture 14
Examination Form	Fachtheoretisch-fachpraktische Arbeit
Examination duration and scale	wird zu Beginn der Veranstaltung festgelegt
Lecturer	Dozenten des SD B
Language	DE/EN
Cycle	WiSe/SoSe
Content	The course occurs only if required. The content is defined at short notice.
Literature	Die Literatur wird kurzfristig festgelegt.

Course L3290: Special topics of the Environmental engineering 2CP	
Typ	
Hrs/wk	2
CP	2
Workload in Hours	Independent Study Time 32, Study Time in Lecture 28
Examination Form	Fachtheoretisch-fachpraktische Arbeit
Examination duration and scale	wird zu Beginn der Veranstaltung festgelegt
Lecturer	Dozenten des SD B
Language	DE/EN
Cycle	WiSe/SoSe
Content	The course occurs only if required. The content is defined at short notice.
Literature	Die Literatur wird kurzfristig festgelegt.

Course L3291: Special topics of the Environmental engineering 3CP	
Typ	
Hrs/wk	3
CP	3
Workload in Hours	Independent Study Time 48, Study Time in Lecture 42
Examination Form	Fachtheoretisch-fachpraktische Arbeit
Examination duration and scale	wird zu Beginn der Veranstaltung festgelegt
Lecturer	Dozenten des SD B
Language	DE/EN
Cycle	WiSe/SoSe
Content	The course occurs only if required. The content is defined at short notice.
Literature	Die Literatur wird kurzfristig festgelegt.

Course L1767: Thermal Biomass Utilization	
Typ	Lecture
Hrs/wk	2
CP	2
Workload in Hours	Independent Study Time 32, Study Time in Lecture 28
Examination Form	Klausur
Examination duration and scale	60 min
Lecturer	Prof. Martin Kaltschmitt
Language	DE
Cycle	WiSe
Content	Goal of this course is it to discuss the physical, chemical, and biological as well as the technical, economic, and environmental basics of all options to provide energy from biomass from a German and international point of view. Additionally different system approaches to use biomass for energy, aspects to integrate bioenergy within the energy system, technical and economic development potentials, and the current and expected future use within the energy system are presented. The course is structured as follows: • Biomass as an energy carrier within the energy system; use of biomass in Germany and world-wide, overview on the content of the course • Photosynthesis, composition of organic matter, plant production, energy crops, residues, organic waste • Biomass provision chains for woody and herbaceous biomass, harvesting and provision, transport, storage, drying • Thermo-chemical conversion of solid biofuels ◦ Basics of thermo-chemical conversion ◦ Direct thermo-chemical conversion through combustion: combustion technologies for small and large scale units, electricity generation technologies, flue gas treatment technologies, ashes and their use ◦ Gasification: Gasification technologies, producer gas cleaning technologies, options to use the cleaned producer gas for the provision of heat, electricity and/or fuels ◦ Fast and slow pyrolysis: Technologies for the provision of bio-oil and/or for the provision of charcoal, oil cleaning technologies, options to use the pyrolysis oil and charcoal as an energy carrier as well as a raw material • Physical-chemical conversion of biomass containing oils and/or fats: Basics, oil seeds and oil fruits, vegetable oil production, production of a biofuel with standardized characteristics (trans-esterification, hydrogenation, co-processing in existing refineries), options to use this fuel, options to use the residues (i.e. meal, glycerine) • Bio-chemical conversion of biomass ◦ Basics of bio-chemical conversion ◦ Biogas: Process technologies for plants using agricultural feedstock, sewage sludge (sewage gas), organic waste fraction (landfill gas), technologies for the provision of bio methane, use of the digested slurry ◦ Ethanol production: Process technologies for feedstock containing sugar, starch or celluloses, use of ethanol as a fuel, use of the stillage
Literature	Kaltschmitt, M.; Hartmann, H. (Hrsg.): Energie aus Biomasse; Springer, Berlin, Heidelberg, 2009, 2. Auflage

Course L2386: Thermal Biomass Utilization	
Typ	Practical Course
Hrs/wk	1
CP	1
Workload in Hours	Independent Study Time 16, Study Time in Lecture 14
Examination Form	Schriftliche Ausarbeitung
Examination duration and scale	Protokolle
Lecturer	Prof. Martin Kaltschmitt, Dr. Marvin Scherzinger
Language	DE
Cycle	WiSe
Content	The experiments of the practical lab course illustrate the different aspects of heat generation from biogenic solid fuels. First, different biomasses (e.g. wood, straw or agricultural residues) will be investigated; the focus will be on the calorific value of the biomass. Furthermore, the used biomass will be pelletized, the pellet properties analysed and a combustion test carried out on a pellet combustion system. The gaseous and solid pollutant emissions, especially the particulate matter emissions, are measured and the composition of the particulate matter is investigated in a further experiment. Another focus of the practical course is the consideration of options for the reduction of particulate matter emissions from biomass combustion. In the practical course, a method for particulate matter reduction will be developed and tested. All experiments will be evaluated and the results presented. Within the practical lab course the students discuss different technical-scientific tasks, both subject-specifically and interdisciplinary. They discuss various approaches to solving the problem and advise on the theoretical or practical implementation.
Literature	- Kaltschmitt, Martin; Hartmann, Hans; Hofbauer, Hermann: Energie aus Biomasse: Grundlagen, Techniken und Verfahren. 3. Auflage. Berlin Heidelberg: Springer Science & Business Media, 2016. -ISBN 978-3-662-47437-2 - Versuchsskript

Module M2032: Advanced Vadose Zone Hydrology				
Courses				
Title	Typ	Hrs/wk	CP	
Modeling Processes in Vadose Zone (L2735)	Recitation Section (small)	2	2	
Vadose Zone Hydrology (L2732)	Lecture	2	2	
Vadose Zone Hydrology (L2733)	Recitation Section (large)	2	2	
Module Responsible	Prof. Nima Shokri			
Admission Requirements	None			
Recommended Previous Knowledge	Basic knowledge in water and soil			
	Comfortable with math and physics, critical thinking, creative problem solving			
	Analytic skills			
Educational Objectives	After taking part successfully, students have reached the following learning results			
Professional Competence				
Knowledge	The students will learn about soil characterization (solid and liquid phase), the energy state of soil water, the soil water characteristic curve, flow in saturated and unsaturated soil as well as about solute transport in soil			
Skills	Students will work on practical examples modelling transport processes in soil using different quantitative tools including computer simulations and analytical tools. This will help them to apply knowledge in order to solve problems and tasks.			
Personal Competence				
Social Competence	The module aims at raising awareness and enthusiasm for new knowledge related to water, soil and environment. This will positively contribute to shape their work and life environment.			
Autonomy	The students will be involved in many problem solving exercises. This will contribute toward their willingness to work independently and responsibly.			
Workload in Hours	Independent Study Time 96, Study Time in Lecture 84			
Credit points	6			
Course achievement	None			
Examination	Written elaboration			
Examination duration and scale	Report and Presentation			
Assignment for the Following Curricula	Civil Engineering: Specialisation Computational Engineering: Elective Compulsory Civil Engineering: Specialisation Water and Traffic: Elective Compulsory Environmental Engineering: Core Qualification: Elective Compulsory Water and Environmental Engineering: Specialisation Water: Elective Compulsory Water and Environmental Engineering: Specialisation Environment: Elective Compulsory			

Course L2735: Modeling Processes in Vadose Zone	
Typ	Recitation Section (small)
Hrs/wk	2
CP	2
Workload in Hours	Independent Study Time 32, Study Time in Lecture 28
Lecturer	Mohammad Aziz Zarif
Language	EN
Cycle	SoSe
Content	Numerical tools will be introduced and used to quantify flow and transport processes in soil
Literature	NA

Course L2732: Vadose Zone Hydrology	
Typ	Lecture
Hrs/wk	2
CP	2
Workload in Hours	Independent Study Time 32, Study Time in Lecture 28
Lecturer	Prof. Nima Shokri
Language	EN
Cycle	SoSe
Content	Soil solid phase characterization, Soil liquid phase characterization, The energy state of soil water, Soil Water Characteristic Curve, Flow in saturated soil, Flow in unsaturated soil, Solute transport in porous media
Literature	- Environmental Soil Physics, by Daniel Hillel - Soil Physics, Sixth Edition, by William A. Jury and Robert Horton - Physical Hydrology, Second Edition, by S. Lawrence Dingman - Introduction to Physical Hydrology, by Martin R. Hendriks

Course L2733: Vadose Zone Hydrology	
Typ	Recitation Section (large)
Hrs/wk	2
CP	2
Workload in Hours	Independent Study Time 32, Study Time in Lecture 28
Lecturer	Sankeerth Govindaiah Narayanaswamy
Language	EN
Cycle	SoSe
Content	See interlocking course
Literature	See interlocking course

Module M1308: Modelling and Technical Design of Bio Refinery Processes			
Courses			
Title	Type	Hrs/wk	CP
Biorefineries - Technical Design and Optimization (L1832)	Project-/problem-based Learning	3	3
CAPE in Energy Engineering (L0022)	Projection Course	3	3
Module Responsible	Prof. Martin Kaltschmitt		
Admission Requirements	None		
Recommended Previous Knowledge	Bachelor degree in Process Engineering, Bioprocess Engineering or Energy- and Environmental Engineering		
Educational Objectives	After taking part successfully, students have reached the following learning results		
Professional Competence <i>Knowledge</i> <i>Skills</i> Personal Competence <i>Social Competence</i> <i>Autonomy</i>	The students can completely design a technical process including mass and energy balances, calculation and layout of different process devices, layout of measurement- and control systems as well as modeling of the overall process. Furthermore, they can describe the basics of the general procedure for the processing of modeling tasks, especially with ASPEN PLUS ® and ASPEN CUSTOM MODELER ®. Students are able to simulate and solve scientific task in the context of renewable energy technologies by: • development of modul-comprehensive approaches for the dimensioning and design of production processes • evaluating alternatives input parameter to solve the particular task even with incomplete information, • a systematic documentation of the work results in form of a written version, the presentation itself and the defense of contents. They can use the ASPEN PLUS ® and ASPEN CUSTOM MODELER ® for modeling energy systems and to evaluate the simulation solutions. Through active discussions of various topics within the seminars and exercises of the module, students improve their understanding and the application of the theoretical background and are thus able to transfer what they have learned in practice. Students can • respectfully work together as a team with around 2-3 members, • participate in subject-specific and interdisciplinary discussions in the area of dimensioning and design of production processes, and can develop cooperated solutions, • defend their own work results in front of fellow students and assess the performance of fellow students in comparison to their own performance. Furthermore, they can accept professional constructive criticism. Students can independently tap knowledge regarding to the given task. They are capable, in consultation with supervisors, to assess their learning level and define further steps on this basis. Furthermore, they can define targets for new application-or research-oriented duties in accordance with the potential social, economic and cultural impact.		
Workload in Hours	Independent Study Time 96, Study Time in Lecture 84		
Credit points	6		
Course achievement	None		
Examination	Written elaboration		
Examination duration and scale	Written report incl. presentation		
Assignment for the Following Curricula	Bioprocess Engineering: Specialisation A - General Bioprocess Engineering: Elective Compulsory Bioprocess Engineering: Specialisation C - Bioeconomic Process Engineering, Focus Energy and Bioprocess Technology: Elective Compulsory Chemical and Bioprocess Engineering: Specialisation Chemical and Bioprocess Engineering: Elective Compulsory Chemical and Bioprocess Engineering: Specialisation Chemical and Bioprocess Engineering: Elective Compulsory Environmental Engineering: Core Qualification: Elective Compulsory Renewable Energies: Core Qualification: Compulsory Process Engineering: Specialisation Environmental Process Engineering: Elective Compulsory		

Course L1832: Biorefineries - Technical Design and Optimization	
Typ	Project-/problem-based Learning
Hrs/wk	3
CP	3
Workload in Hours	Independent Study Time 48, Study Time in Lecture 42
Lecturer	Prof. Oliver Lüdtkke
Language	DE
Cycle	SoSe
Content	I. Repetition of engineering basics 1. Shell and tube heat exchangers 2. Steam generators and refrigerating machines 3. Pumps and turbines 4. Flow in piping networks 5. Pumping and mixing of non-newtonian fluids 6. Requirements to a detailed layout plan II. Calculation: 1. Planning and design of a specific bio-refinery plant section, such as Ethanol distillation and fermentation. This is based on empirical value of a real, industrial plant. ◦ Mass and energy balances (Aspen) ◦ Equipment design (heat exchangers, pumps, pipes, tanks, etc.) (◦ Isolation, wall thickness and material selection ◦ Energy demand (electrical, heat or cooling), design of steam boilers and appliances ◦ Selection of fittings, measuring instruments and safety equipment ◦ Definition of main control loops 2. Hereby, the dependencies of transport phenomena between certain plant sections become evident and methods of calculation are introduced. 3. In Detail Engineering , it is focused on aspects of plant engineering planning that are relevant for the subsequent construction of the plant. 4. Depending of time requirement and group size a cost estimation and preparation of a complete R&I flow chart can be implemented as well.
Literature	Perry, R.;Green, R.: Perry's Chemical Engineers' Handbook, 8th Edition, McGraw Hill Professional, 2007 Sinnot, R. K.: Chemical Engineering Design, Elsevier, 2014

Course L0022: CAPE in Energy Engineering	
Typ	Projection Course
Hrs/wk	3
CP	3
Workload in Hours	Independent Study Time 48, Study Time in Lecture 42
Lecturer	Prof. Martin Kaltschmitt
Language	DE
Cycle	SoSe
Content	• CAPE = <i>Computer-Aided-Project-Engineering</i> • INTRODUCTION TO THE THEORY ◦ Classes of simulation programs ◦ Sequential modular approach ◦ Equation-oriented approach ◦ Simultaneous modular approach ◦ General procedure for the processing of modeling tasks ◦ Special procedure for solving models with repatriations • COMPUTER EXERCISES renewable energy projects WITH ASPEN PLUS ® AND ASPEN CUSTOM MODELER ® ◦ Scope, potential and limitations of Aspen Plus ® and Aspen Custom Modeler ® ◦ Use of integrated databases for material data ◦ Methods for estimating non-existent physical property data ◦ Use of model libraries and Process Synthesis ◦ Application of design specifications and sensitivity analyzes ◦ Solving optimization problems Within the seminar, the various tasks are actively discussed and applied to various cases of application.
Literature	• Aspen Plus® - Aspen Plus User Guide • William L. Luyben; Distillation Design and Control Using Aspen Simulation; ISBN-10: 0-471-77888-5

Module M2155: Uncertainty Modelling for Engineers				
Courses				
Title	Typ		Hrs/wk	CP
Uncertainty Modelling for Engineers (L3458)	Integrated Lecture		6	6
Module Responsible	Prof. Simon Michael Papalexiou			
Admission Requirements	None			
Recommended Previous Knowledge	1. General familiarity with engineering concepts. 2. Elementary probability and statistics, and mathematical skills. 3. Basic computer skills for handling data. 4. Interest in solving engineering problems using statistical and probabilistic methods.			
Educational Objectives	After taking part successfully, students have reached the following learning results			
Professional Competence	<i>Knowledge</i> Students will develop a strong foundation in uncertainty, probability, and risk analysis in engineering applications. The course introduces probability as a measure of uncertainty, covering frequency-based methods. Students will explore Bayes' Theorem, probability distributions, extreme value theory, joint probability distributions, and stochastic optimization to model and quantify uncertainty in engineering problems. The course also covers linear and nonlinear regression methods, essential for data-driven decision-making and predictive modeling. Additionally, students will gain insight into risk assessment as a function of probability and disutility and learn how to apply Bayesian Decision Theory to optimize engineering solutions under uncertainty. <i>Skills</i> By the end of the course, students will be able to apply probabilistic models to quantify uncertainty and assess risks in engineering problems. They will gain expertise in fitting probability distributions, performing extreme value analysis, and applying Bayesian inference to real-world engineering challenges. Students will also develop skills in linear and nonlinear regression modeling, enabling them to analyze complex engineering datasets and improve risk predictions. Through hands-on computational exercises, they will learn to implement stochastic methods and optimization techniques to support reliability-based design and engineering decision-making. <i>Social Competence</i> Students will develop the ability to work collaboratively on engineering risk assessments, communicating technical results effectively with peers, engineers, and decision-makers. They will engage in discussions on risk perception, safety factors, and uncertainty quantification, ensuring that engineering analyses are both rigorous and applicable to real-world infrastructure challenges. <i>Autonomy</i> Students will learn to independently analyze and model engineering uncertainties, selecting and applying appropriate probability distributions, regression methods, and stochastic techniques for various applications. They will also gain the ability to evaluate risks associated with natural and human-made hazards, ensuring they can make informed engineering decisions in design, safety assessment, and disaster mitigation.			
Workload in Hours				
Credit points				
Course achievement	Compulsory	Bonus	Form	Description
	Yes	20 %	Presentation	10-minütige Präsentation von Arbeitsergebnissen
Examination	Written exam			
Examination duration and scale	150 min			
Assignment for the Following Curricula	Civil Engineering: Specialisation Coastal Engineering: Elective Compulsory Civil Engineering: Specialisation Geotechnical Engineering: Elective Compulsory Civil Engineering: Specialisation Structural Engineering: Elective Compulsory Civil Engineering: Specialisation Computational Engineering: Elective Compulsory Civil Engineering: Specialisation Water and Traffic: Elective Compulsory Civil Engineering: Specialisation Coastal Engineering: Elective Compulsory Civil Engineering: Specialisation Geotechnical Engineering: Elective Compulsory Civil Engineering: Specialisation Structural Engineering: Elective Compulsory Civil Engineering: Specialisation Computational Engineering: Elective Compulsory Civil Engineering: Specialisation Water and Traffic: Elective Compulsory Environmental Engineering: Core Qualification: Elective Compulsory Environmental Engineering: Core Qualification: Elective Compulsory Water and Environmental Engineering: Specialisation Cities: Elective Compulsory Water and Environmental Engineering: Specialisation Environment: Elective Compulsory Water and Environmental Engineering: Specialisation Water: Elective Compulsory Water and Environmental Engineering: Specialisation Cities: Elective Compulsory Water and Environmental Engineering: Specialisation Environment: Elective Compulsory Water and Environmental Engineering: Specialisation Water: Elective Compulsory			

Course L3458: Uncertainty Modelling for Engineers	
Type	Integrated Lecture
Hrs/wk	6
CP	6
Workload in Hours	Independent Study Time 96, Study Time in Lecture 84
Lecturer	Prof. Simon Michael Papalexiou
Language	EN
Cycle	SoSe
Content	Engineering decisions are rarely made with complete certainty—uncertainty affects material properties, environmental conditions, structural performance, and risk assessments. This course provides students with theoretical foundations and practical tools to quantify uncertainty, assess risks, and enhance decision-making in civil, structural, geotechnical, and environmental engineering applications. Students will begin with fundamental probability concepts, learning how Bayes' Theorem, probability distributions, and extreme value theory help evaluate engineering uncertainties. They will explore linear and nonlinear regression methods for analyzing complex datasets, as well as joint probability distributions and stochastic optimization to improve predictive modeling and reliability assessments. The course also introduces Bayesian Decision Theory, offering a structured approach to decision-making under uncertainty. With a focus on real-world engineering problems, students will apply probabilistic models, extreme value analysis, and stochastic techniques to assess risks in infrastructure design, system reliability, and disaster resilience. Hands-on computational exercises will reinforce key concepts, preparing students to work with data-driven models and uncertainty quantification techniques used in engineering practice. This course is ideal for students interested in engineering risk assessment, reliability analysis, and data-driven modeling. By the end of the course, students will have developed critical analytical and problem-solving skills, equipping them for careers in structural safety, geotechnical engineering, environmental risk management, and beyond.
Literature	

Module M2076: Introduction to Climate Informed Engineering			
Courses			
Title	Type	Hrs/wk	CP
Methods in Climate Informed Engineering (L3347)	Lecture	3	3
Topics in Climate Informed Engineering (L3348)	Lecture	3	3
Module Responsible	Prof. Nima Shokri		
Admission Requirements	None		
Recommended Previous Knowledge	Students are expected to have a foundational understanding of environmental science, basic engineering principles, and an interest in sustainability. Recommended knowledge includes climate science, data analysis, and familiarity with engineering design processes. Analytical and critical thinking and creative problem-solving skills are also beneficial		
Educational Objectives	After taking part successfully, students have reached the following learning results		
Professional Competence <i>Knowledge</i>	This module explores next-generation climate models and high-resolution data, emphasizing their impact on environmental and engineering products and processes. It covers how various engineering disciplines can benefit from climate information. Research-based learning activities, expert talks, and presentations will expose students to state-of-the-art modeling, measurement, and analysis in climate-informed engineering.		
<i>Skills</i>			
Personal Competence <i>Social Competence</i>			
<i>Autonomy</i>			
Workload in Hours	Independent Study Time 96, Study Time in Lecture 84		
Credit points	6		
Course achievement	None		
Examination	Subject theoretical and practical work		
Examination duration and scale	Report and Presentation		
Assignment for the Following Curricula	Civil Engineering: Specialisation Coastal Engineering: Elective Compulsory Civil Engineering: Specialisation Geotechnical Engineering: Elective Compulsory Civil Engineering: Specialisation Structural Engineering: Elective Compulsory Civil Engineering: Specialisation Water and Traffic: Elective Compulsory Civil Engineering: Specialisation Computational Engineering: Elective Compulsory Data Science: Specialisation III. Applications: Elective Compulsory Environmental Engineering: Core Qualification: Elective Compulsory Process Engineering: Specialisation Process Engineering: Elective Compulsory Water and Environmental Engineering: Specialisation Cities: Elective Compulsory Water and Environmental Engineering: Specialisation Environment: Elective Compulsory Water and Environmental Engineering: Specialisation Water: Elective Compulsory		

Course L3347: Methods in Climate Informed Engineering	
Type	Lecture
Hrs/wk	3
CP	3
Workload in Hours	Independent Study Time 48, Study Time in Lecture 42
Lecturer	Prof. Nima Shokri, Prof. Cathy Hohenegger, Prof. Irina Smirnova
Language	EN
Cycle	WiSe
Content	Students will learn techniques for incorporating climate data and environmental factors into engineering design. It covers climate modelling and the use of sensors and devices to measure climate-related parameters and engineering processes. Students will have the opportunity to conduct their own measurements, analyze the collected data, and write a report on their findings. This hands-on experience will be assessed and contribute to their final grade.
Literature	

Course L3348: Topics in Climate Informed Engineering	
Typ	Lecture
Hrs/wk	3
CP	3
Workload in Hours	Independent Study Time 48, Study Time in Lecture 42
Lecturer	Prof. Irina Smirnova, Prof. Cathy Hohenegger, Prof. Nima Shokri
Language	EN
Cycle	WiSe
Content	Exploring specific applications of climate data in various engineering disciplines. Invited speakers will present their research and discuss the relevance of climate-informed engineering to their work. Additionally, there will be a segment on effective communication, covering how to give impactful presentations and write research papers. Students will also give presentations on their own class projects related to climate-informed engineering, applying the concepts they've learned. This hands-on experience will be assessed and contribute to their final grade.
Literature	

Specialization Energy and Resources

Module M1724: Smart Monitoring

Courses

Title	Typ	Hrs/wk	CP
Smart Monitoring (L2762)	Integrated Lecture	2	2
Smart Monitoring (L2763)	Recitation Section (small)	2	4
Module Responsible	Prof. Kay Smarsly		
Admission Requirements	None		
Recommended Previous Knowledge	Basic knowledge or interest in object-oriented modeling, programming, and sensor technologies are helpful. Interest in modern research and teaching areas, such as Internet of Things, Industry 4.0 and cyber-physical systems, as well as the will to deepen skills of scientific working, are required. Basic knowledge in scientific writing and good English skills.		
Educational Objectives	After taking part successfully, students have reached the following learning results		
Professional Competence	The students will become familiar with the principles and practices of smart monitoring. The students will be able to design decentralized smart systems to be applied for continuous (remote) monitoring of systems in the built and in the natural environment. In addition, the students will learn to design and to implement intelligent sensor systems using state-of-the-art data analysis techniques, modern software design concepts, and embedded computing methodologies. Besides lectures, project work is also part of this module, which will be conducted throughout the semester and will contribute to the grade. In small groups, the students will design smart monitoring systems that integrate a number of "intelligent" sensors to be implemented by the students. Specific focus will be put on the application of machine learning techniques. The smart monitoring systems will be mounted on real-world (built or natural) systems, such as bridges or slopes, or on scaled lab structures for validation purposes. The outcome of every group will be documented in a paper. All students of this module will "automatically" participate with their smart monitoring system in the annual "Smart Monitoring" competition. The written papers and oral examinations form the final grades. The module will be taught in English. Limited enrollment. The students will gain insights into operating state-of-the-art smart sensor systems, used for monitoring a wide range of physical processes relevant to engineering, such as environmental, structural, or comfort monitoring. The students will be capable of devising monitoring strategies of physical processes as part of group projects, tailored to their knowledge backgrounds, and to implement the strategies in smart wireless sensor nodes, using embedded computing and programming. Finally, the students will be able to document the findings of their projects in short reports. The students will be able to work in groups, share parts of the work for their projects, and develop communication skills, towards achieving the common project goals. The students will be able to gain a solid basis on approaching and solving problems in engineering, as well as on documenting results, through their involvement in their monitoring group projects.		
<i>Knowledge</i>			
<i>Skills</i>			
Personal Competence			
<i>Social Competence</i>			
<i>Autonomy</i>			
Workload in Hours	Independent Study Time 124, Study Time in Lecture 56		
Credit points	6		
Course achievement	None		
Examination	Written elaboration		
Examination duration and scale	10 pages of work with 15-minute oral presentation		
Assignment for the Following Curricula	Civil Engineering: Specialisation Water and Traffic: Elective Compulsory Civil Engineering: Specialisation Geotechnical Engineering: Elective Compulsory Civil Engineering: Specialisation Coastal Engineering: Elective Compulsory Civil Engineering: Specialisation Structural Engineering: Elective Compulsory Computer Science: Specialisation II: Intelligence Engineering: Elective Compulsory Environmental Engineering: Specialisation Energy and Resources: Elective Compulsory Environmental Engineering: Specialisation Environment and Climate: Elective Compulsory Environmental Engineering: Specialisation Water Quality and Water Engineering: Elective Compulsory Mechatronics: Technical Complementary Course: Elective Compulsory Mechatronics: Core Qualification: Elective Compulsory Theoretical Mechanical Engineering: Specialisation Robotics and Computer Science: Elective Compulsory Water and Environmental Engineering: Specialisation Cities: Elective Compulsory Water and Environmental Engineering: Specialisation Environment: Elective Compulsory Water and Environmental Engineering: Specialisation Water: Elective Compulsory		

Course L2762: Smart Monitoring	
Typ	Integrated Lecture
Hrs/wk	2
CP	2
Workload in Hours	Independent Study Time 32, Study Time in Lecture 28
Lecturer	Prof. Kay Smarsly
Language	EN
Cycle	SoSe
Content	In this course, principles of smart monitoring will be taught, focusing on modern concepts of data acquisition, data storage, and data analysis. Also, fundamentals of intelligent sensors and embedded computing will be illuminated. Autonomous software and decentralized data processing are further crucial parts of the course, including concepts of the Internet of Things, Industry 4.0 and cyber-physical systems. Furthermore, measuring principles, data acquisition systems, data management and data analysis algorithms will be discussed. Besides the theoretical background, numerous practical examples will be shown to demonstrate how smart monitoring may advantageously be used for assessing the condition of systems in the built or natural environment.
Literature	The course contents couples different fields, such as signal processing, sensing technologies, data analytics, environmental engineering, civil engineering, artificial intelligence, database systems, and many more. The basics will be taught in this course. However, specific literature that covers all these topics does not exist. Instead, literature will be referenced in the lectures, all of which are papers that are freely available online.

Course L2763: Smart Monitoring	
Typ	Recitation Section (small)
Hrs/wk	2
CP	4
Workload in Hours	Independent Study Time 92, Study Time in Lecture 28
Lecturer	Prof. Kay Smarsly
Language	EN
Cycle	SoSe
Content	The contents of the exercises are based on the lecture contents. In addition to the exercises, project work will be conducted throughout the semester, which will consume the majority of the workload. As part of the project work, students will design smart monitoring systems that will be tested in the laboratory or in the field. As mentioned in the module description, the students will participate in the "Smart Monitoring" competition, hosted annually by the Institute of Digital and Autonomous Construction. Students are encouraged to contribute their own ideas. The tools required to implement the smart monitoring systems will be taught in the group exercises as well as through external sources, such as video tutorials and literature.
Literature	The course contents couples different fields, such as signal processing, sensing technologies, data analytics, environmental engineering, civil engineering, artificial intelligence, database systems, and many more. The basics will be taught in this course. However, specific literature that covers all these topics does not exist. Instead, literature will be referenced in the lectures, all of which are papers that are freely available online.

Module M1709: Applied Optimization in Energy and Process Engineering				
Courses				
Title	Typ		Hrs/wk	CP
Applied optimization in energy and process engineering (L2693)	Integrated Lecture		2	3
Applied optimization in energy and process engineering (L2695)	Recitation Section (small)		3	3
Module Responsible	Prof. Mirko Skiborowski			
Admission Requirements	None			
Recommended Previous Knowledge	Fundamentals in the field of mathematical modeling and numerical mathematics, as well as a basic understanding of process engineering processes. In particular the contents of the module Process and Plant Engineering II			
Educational Objectives	After taking part successfully, students have reached the following learning results			
Professional Competence <i>Knowledge</i>	The module provides a general introduction to the basics of applied mathematical optimization and deals with application areas on different scales from the identification of kinetic models, to the optimal design of unit operations and the optimization of entire (sub)processes, as well as production planning. In addition to the basic classification and formulation of optimization problems, different solution approaches are discussed and tested during the exercises. Besides deterministic gradient-based methods, metaheuristics such as evolutionary and genetic algorithms and their application are discussed as well. • Introduction to Applied Optimization • Formulation of optimization problems • Linear Optimization • Nonlinear Optimization • Mixed-integer (non)linear optimization • Multi-objective optimization • Global optimization			
<i>Skills</i>	After successful participation in the module "Applied Optimization in Energy and Process Engineering", students are able to formulate the different types of optimization problems and to select appropriate solution methods in suitable software such as Matlab and GAMS and to develop improved solution strategies. Furthermore, students will be able to interpret and critically examine the results accordingly.			
Personal Competence <i>Social Competence</i>	Students are capable of: • develop solutions in heterogeneous small groups			
<i>Autonomy</i>	Students are capable of: • tapping new knowledge on a special subject by literature research			
Workload in Hours	Independent Study Time 110, Study Time in Lecture 70			
Credit points	6			
Course achievement	Compulsory	Bonus	Form	Description
	No	10 %	Midterm	Bonuspunkte
Examination	Oral exam			
Examination duration and scale	35 min			
Assignment for the Following Curricula	Bioprocess Engineering: Specialisation A - General Bioprocess Engineering: Elective Compulsory Chemical and Bioprocess Engineering: Specialisation Chemical and Bioprocess Engineering: Elective Compulsory Chemical and Bioprocess Engineering: Specialisation Chemical and Bioprocess Engineering: Elective Compulsory Computational Engineering: Core Qualification: Elective Compulsory Energy Systems: Specialisation Energy Systems: Elective Compulsory Environmental Engineering: Specialisation Energy and Resources: Elective Compulsory Renewable Energies: Specialisation Bioenergy Systems: Elective Compulsory Renewable Energies: Specialisation Wind Energy Systems: Elective Compulsory Technomathematics: Specialisation III. Engineering Science: Elective Compulsory Theoretical Mechanical Engineering: Specialisation Energy Systems: Elective Compulsory Process Engineering: Specialisation Chemical Process Engineering: Elective Compulsory Process Engineering: Specialisation Process Engineering: Elective Compulsory			

Course L2693: Applied optimization in energy and process engineering	
Typ	Integrated Lecture
Hrs/wk	2
CP	3
Workload in Hours	Independent Study Time 62, Study Time in Lecture 28
Lecturer	Prof. Mirko Skiborowski
Language	EN
Cycle	SoSe
Content	The lecture offers a general introduction to the basics and possibilities of applied mathematical optimization and deals with application areas on different scales from kinetics identification, optimal design of unit operations to the optimization of entire (sub)processes, and production planning. In addition to the basic classification and formulation of optimization problems, different solution approaches are discussed. Besides deterministic gradient-based methods, metaheuristics such as evolutionary and genetic algorithms and their application are discussed as well. - Introduction to Applied Optimization - Formulation of optimization problems - Linear Optimization - Nonlinear Optimization - Mixed-integer (non)linear optimization - Multi-objective optimization - Global optimization
Literature	Weicker, K., Evolutionäre Algorithmen, Springer, 2015 Edgar, T. F., Himmelblau D. M., Lasdon, L. S., Optimization of Chemical Processes, McGraw Hill, 2001 Biegler, L. Nonlinear Programming - Concepts, Algorithms, and Applications to Chemical Processes, 2010 Kallrath, J. Gemischt-ganzzahlige Optimierung: Modellierung in der Praxis, Vieweg, 2002

Course L2695: Applied optimization in energy and process engineering	
Typ	Recitation Section (small)
Hrs/wk	3
CP	3
Workload in Hours	Independent Study Time 48, Study Time in Lecture 42
Lecturer	Prof. Mirko Skiborowski
Language	EN
Cycle	SoSe
Content	See interlocking course
Literature	See interlocking course

Module M2002: Waste and Resource Management				
Courses				
Title	Typ		Hrs/wk	CP
Waste management (L3261)	Project-/problem-based Learning		3	3
International waste concepts (L3259)	Lecture		2	2
International waste concepts (L3260)	Recitation Section (small)		1	1
Module Responsible	Prof. Kerstin Kuchta			
Admission Requirements	None			
Recommended Previous Knowledge	Basics in process engineering			
Educational Objectives	After taking part successfully, students have reached the following learning results			
Professional Competence	<i>Knowledge</i> The students are able to describe waste as a resource as well as advanced technologies for recycling and recovery of resources from waste in detail. This covers collection, transport, treatment and disposal in national and international contexts. <i>Skills</i> Students are able to select suitable processes for the treatment with respect to the national or cultural and developmental context. They can evaluate the ecological impact and the technical effort of different technologies and management systems. Personal Competence <i>Social Competence</i> Students can work together as a team of 2-5 persons, participate in subject-specific and interdisciplinary discussions, develop cooperated solutions and defend their own work results in front of others and promote the scientific development of colleagues. Furthermore, they can give and accept professional constructive criticisms. <i>Autonomy</i> Students can independently gain additional knowledge of the subject area and apply it in solving the given course tasks and projects.			
Workload in Hours	Independent Study Time 96, Study Time in Lecture 84			
Credit points	6			
Course achievement	Compulsory	Bonus	Form	Description
	Yes	20 %	Written elaboration	
Examination	Presentation			
Examination duration and scale	PowerPoint presentation (10-15 minutes)			
Assignment for the Following Curricula	Civil Engineering: Specialisation Water and Traffic: Elective Compulsory Chemical and Bioprocess Engineering: Specialisation Chemical and Bioprocess Engineering: Elective Compulsory Chemical and Bioprocess Engineering: Specialisation Chemical and Bioprocess Engineering: Elective Compulsory Chemical and Bioprocess Engineering: Core Qualification: Elective Compulsory Environmental Engineering: Specialisation Energy and Resources: Elective Compulsory International Management and Engineering: Specialisation II. Renewable Energy: Elective Compulsory Process Engineering: Specialisation Environmental Process Engineering: Elective Compulsory Water and Environmental Engineering: Specialisation Cities: Elective Compulsory Water and Environmental Engineering: Specialisation Environment: Elective Compulsory			

Course L3261: Waste management	
Typ	Project-/problem-based Learning
Hrs/wk	3
CP	3
Workload in Hours	Independent Study Time 48, Study Time in Lecture 42
Lecturer	Prof. Rüdiger Siechau
Language	EN
Cycle	SoSe
Content	• Introduction into the "Waste Management" consisting of: ◦ Thermal Process (incinerator, RDF combustion) ◦ Biological processes (Wet-/Dryfermentation) ◦ technology, energy, emissions, approval , etc. • Group work ◦ design of systems/plants for energy recovery from waste ◦ The following points are to be processed: ▪ Input: waste (fraction collection and transportation, current quantity, material flows , possible amount of development) ▪ Plant (design, process diagram, technology, energy production) ▪ Output (energy quantity / type, by-products) ▪ Costs and revenues ▪ Climate and resource protection (CO2 balance , substitution of primary raw materials / fossil fuels) ▪ Location and approval (infrastructure , expiration authorization procedure) ▪ Focus at the whole concept (advantages, disadvantages , risks and opportunities , discussion)
Literature	Einführung in die Abfallwirtschaft; Martin Kranert, Klaus Cord-Landwehr (Hrsg.); Vieweg + Teubner Verlag; 2010 Powerpoint-Folien in Stud IP

Course L3259: International waste concepts	
Typ	Lecture
Hrs/wk	2
CP	2
Workload in Hours	Independent Study Time 32, Study Time in Lecture 28
Lecturer	Prof. Kerstin Kuchta
Language	EN
Cycle	SoSe
Content	Waste avoidance and recycling are the focus of this lecture. Additionally, waste logistics (Collection, transport, export, fees and taxes) as well as international waste shipment solutions are presented. Other specific wastes, e.g. industrial waste, treatment concepts will be presented and developed by students themselves Waste composition and production on international level, waste eulogistic, collection and treatment in emerging and developing countries. Single national projects and studies will be prepared and presented by students
Literature	Basel convention

Course L3260: International waste concepts	
Typ	Recitation Section (small)
Hrs/wk	1
CP	1
Workload in Hours	Independent Study Time 16, Study Time in Lecture 14
Lecturer	Prof. Kerstin Kuchta
Language	EN
Cycle	SoSe
Content	See interlocking course
Literature	See interlocking course

Module M1899: Study work Energy and Resources				
Courses				
Title	Typ		Hrs/wk	CP
Module Responsible	Dozenten des Studiengangs			
Admission Requirements	None			
Recommended Previous Knowledge				
Educational Objectives	After taking part successfully, students have reached the following learning results			
Professional Competence <i>Knowledge</i>	The students can demonstrate their detailed knowledge in an area of energy and resource management. The students are qualified to project energy technology and especially resource technology projects and to independently define research tasks for the theoretical and experimental investigation of material and energy issues. They are able to give examples of the state of development and application and to discuss these critically, taking into account current problems and framework conditions in science and society. The students are able to independently define a solution strategy for a basic, application-oriented or practical problem from the field of resource and energy technology and to outline individual solution approaches. In doing so, they are able to proceed in a theory-oriented manner and include current safety, ecological, ethical and economic aspects according to the state of the art in science and associated social discussions. They can use the scientific working techniques they have chosen for their own project work, they can present them in detail and critically discuss them.			
<i>Skills</i>	Students are able to independently select methodological approaches for project work and justify this selection in terms of content. They can explain how they relate approaches or methods to the specific field of application in a solution-oriented manner and adapt them to the application context. They can outline the main points and further developments that go beyond the project.			
Personal Competence <i>Social Competence</i>	Students are able to prepare the relevance and cut of their project task, the work steps and sub-problems for discussion and debate in larger groups, guide the discussions and give feedback to colleagues on their projects.			
<i>Autonomy</i>	The students are able to independently plan and document the work steps and processes necessary to complete the coursework, taking into account specified deadlines. This includes being able to obtain current scientific information in a goal-oriented manner. Furthermore, they are able to obtain feedback on the progress of work from experts in the field in order to achieve high-quality work results based on the state of the art in science and technology.			
Workload in Hours	Independent Study Time 360, Study Time in Lecture 0			
Credit points	12			
Course achievement	None			
Examination	Study work			
Examination duration and scale	depending on task			
Assignment for the Following Curricula	Environmental Engineering: Specialisation Energy and Resources: Compulsory			

Module M1354: Advanced Fuels				
Courses				
Title		Type	Hrs/wk	CP
Second generation biofuels and electricity based fuels (L2414)		Lecture	2	2
Carbon dioxide as an economic determinant in the mobility sector (L1926)		Lecture	1	1
Mobility and climate protection (L2416)		Recitation Section (small)	2	2
Sustainability aspects and regulatory framework (L2415)		Lecture	1	1
Module Responsible	Prof. Martin Kaltschmitt			
Admission Requirements	None			
Recommended Previous Knowledge	Bachelor degree in Process Engineering, Bioprocess Engineering or Energy- and Environmental Engineering			
Educational Objectives	After taking part successfully, students have reached the following learning results			
Professional Competence				
<i>Knowledge</i>				
<i>Skills</i>				
Personal Competence				
<i>Social Competence</i>				
<i>Autonomy</i>				
Workload in Hours	Independent Study Time 96, Study Time in Lecture 84			
Credit points	6			
Course achievement	Compulsory	Bonus	Form	Description
	Yes	20 %	Written elaboration	Details werden in der ersten Veranstaltung bekannt gegeben.
Examination	Written exam			
Examination duration and scale	120 min			
Assignment for the Following Curricula	Bioprocess Engineering: Specialisation A - General Bioprocess Engineering: Elective Compulsory Bioprocess Engineering: Specialisation B - Industrial Bioprocess Engineering: Elective Compulsory Bioprocess Engineering: Specialisation C - Bioeconomic Process Engineering, Focus Energy and Bioprocess Technology: Elective Compulsory Chemical and Bioprocess Engineering: Specialisation Chemical and Bioprocess Engineering: Elective Compulsory Chemical and Bioprocess Engineering: Specialisation Chemical and Bioprocess Engineering: Elective Compulsory Energy Systems: Specialisation Energy Systems: Elective Compulsory Environmental Engineering: Specialisation Energy and Resources: Elective Compulsory Aircraft Systems Engineering: Core Qualification: Elective Compulsory Logistics, Infrastructure and Mobility: Specialisation Production and Logistics: Elective Compulsory Logistics, Infrastructure and Mobility: Specialisation Infrastructure and Mobility: Elective Compulsory Renewable Energies: Specialisation Wind Energy Systems: Elective Compulsory Renewable Energies: Specialisation Solar Energy Systems: Elective Compulsory Renewable Energies: Specialisation Bioenergy Systems: Elective Compulsory Process Engineering: Specialisation Process Engineering: Elective Compulsory Process Engineering: Specialisation Chemical Process Engineering: Elective Compulsory Process Engineering: Specialisation Environmental Process Engineering: Elective Compulsory			

Course L2414: Second generation biofuels and electricity based fuels	
Typ	Lecture
Hrs/wk	2
CP	2
Workload in Hours	Independent Study Time 32, Study Time in Lecture 28
Lecturer	Prof. Martin Kaltschmitt
Language	DE/EN
Cycle	WiSe
Content	- General overview of various power-based fuels and their process paths, including power-to-liquid process (Fischer-Tropsch synthesis, methanol synthesis), power-to-gas (Sabatier process) - Origin, production and use of these fuels
Literature	Vorlesungsskript

Course L1926: Carbon dioxide as an economic determinant in the mobility sector	
Typ	Lecture
Hrs/wk	1
CP	1
Workload in Hours	Independent Study Time 16, Study Time in Lecture 14
Lecturer	Dr. Karsten Wilbrand
Language	DE/EN
Cycle	WiSe
Content	- General overview of various advanced biofuels and their process pathways (including gas-to-liquid, HEFA and Alcohol-to-Jet processes) - Origin, production and use of these fuels
Literature	- Babu, V.: Biofuels Production. Beverly, Mass: Scrivener [u.a.], 2013 - Olsson, L.: Biofuels. Berlin, Heidelberg: Springer-Verlag Berlin Heidelberg, 2007 - William, L. L.: Distillation Design and Control Using Aspen Simulation; ISBN-10: 0-471-77888-5 - Perry, R.; Green, R.: Perry's Chemical Engineers' Handbook, 8th Edition, McGraw Hill Professional, 20 - Sinnot, R. K.: Chemical Engineering Design, Elsevier, 2014 - Kaltschmitt, M.; Neuling, U. (Ed.): Biokerosene - Status and Prospects; Springer, Berlin, Heidelberg, 2018

Course L2416: Mobility and climate protection	
Typ	Recitation Section (small)
Hrs/wk	2
CP	2
Workload in Hours	Independent Study Time 32, Study Time in Lecture 28
Lecturer	Dr. Benedikt Buchspies, Dr. Karsten Wilbrand
Language	DE/EN
Cycle	WiSe
Content	Application of the acquired theoretical knowledge from the respective lectures on the basis of concrete tasks from practice - Design and simulation of sub-processes of production processes in Aspen Plus ® - Ecological and economic analysis of fuel supply paths - Classification of case studies into applicable regulations
Literature	- Skriptum zur Vorlesung - Aspen Plus® - Aspen Plus User Guide

Course L2415: Sustainability aspects and regulatory framework	
Typ	Lecture
Hrs/wk	1
CP	1
Workload in Hours	Independent Study Time 16, Study Time in Lecture 14
Lecturer	Dr. Benedikt Buchspies
Language	DE/EN
Cycle	WiSe
Content	Holistic examination of the different fuel paths with the following main topics, among others: • Consideration of the environmental impact of the various alternative fuels • Economic consideration of the different alternative fuels • Regulatory framework for alternative fuels • Certification of alternative fuels • Market introduction models of alternative fuels
Literature	• European Commission - Joint Research Center (2010): International Reference Life Cycle Data System (ILCD) Handbook - General guide for Life Cycle Assessment - Detailed guidance. Joint Research Center (JRC) Institut for Environment and Sustainability, Luxembourg • Richtlinie (EU) 2018/2001 des Europäischen Parlaments und des Rates vom 11. Dezember 2018 zur Förderung der Nutzung von Energie aus erneuerbaren Quellen

Module M1125: Bioresources and Biorefineries			
Courses			
Title	Type	Hrs/wk	CP
Biorefinery Technology (L0895)	Lecture	2	2
Biorefinery Technologie (L0974)	Recitation Section (small)	1	1
Bioresource Management (L0892)	Lecture	2	2
Bioresource Management (L0893)	Recitation Section (small)	1	1
Module Responsible	Dr. Ina Körner		
Admission Requirements	None		
Recommended Previous Knowledge	Basics on engineering; Basics of waste and energy management		
Educational Objectives	After taking part successfully, students have reached the following learning results		
Professional Competence	<i>Knowledge</i> Students can give an overview on principles and theories in the field's bioresource management and biorefinery technology and can explain specialized terms and technologies. <i>Skills</i> Students are capable of applying knowledge and know-how in the field's bioresource management and biorefinery technology in order to perform technical and regional-planning tasks. They are also able to discuss the links to waste management, energy management and biotechnology. Personal Competence <i>Social Competence</i> Students can work goal-oriented with others and communicate and document their interests and knowledge in acceptable way. <i>Autonomy</i> Students are able to solve independently, with the aid of pointers, practice-related tasks bearing in mind possible societal consequences.		
Workload in Hours	Independent Study Time 96, Study Time in Lecture 84		
Credit points	6		
Course achievement	None		
Examination	Written exam		
Examination duration and scale	90 min		
Assignment for the Following Curricula	Chemical and Bioprocess Engineering: Specialisation Bioprocess Engineering: Elective Compulsory Environmental Engineering: Specialisation Energy and Resources: Elective Compulsory International Management and Engineering: Specialisation II. Energy and Environmental Engineering: Elective Compulsory		

Course L0895: Biorefinery Technology	
Typ	Lecture
Hrs/wk	2
CP	2
Workload in Hours	Independent Study Time 32, Study Time in Lecture 28
Lecturer	Dr. Ina Körner
Language	EN
Cycle	WiSe
Content	The Europe 2020 strategy calls for bioeconomy as the key for smart and green growth of today. Biorefineries are the fundamental part on the way to convert the use of fossil-based society to bio-based society. For this reason, agriculture and forestry sectors are increasingly deliver bioresources. It is not only for their traditional applications in the food and feed sectors such as pulp or paper and construction material productions, but also to produce bioenergy and bio-based products such as bio-plastics. However although bioresources are renewable, they are considered as limited resources as well. The bioeconomy's limitation factor is the availability land on our world. In the context of the development of the bioeconomy, the sustainable and reliable supply of non-food biomass feedstock is a critical success factor for the long-term perspective of bioenergy and other bio-based products production. Biorefineries are complex of technologies and process cascades using the available primary, secondary and tertiary bioresources to produce a multitude of products - a product mix from material and energy products. The lecture gives an overview on biorefinery technology and shall contribute to promotion of international biorefinery developments. Lectures: • What is a biorefinery: Overview on basic organic substrates and processes which lead to material and energy products • The way from a fossil based to a biobased economy in the 21st century • The worlds most advanced biorefinery • Presentation of various biorefinery systems and their products (e.g. lignocellulose biorefinery, green biorefinery, whole plant biorefinery, civilization biorefinery) • Example projects (e.g. combination of anaerobic digestion and composting in practice; demonstration project in Hamburgs city quarter Jenfelder Au) The lectures will be accompanied by technical tours. Optional it is also possible to visit more biorefinery lectures in the University of Hamburg (lectures in German only). In the exercise students have the possibility to work in groups on a biorefinery project or to work on a student-specific task.
Literature	Biorefineries - Industrial Process and Products - Status Qua and Future directions by Kamm, Gruber and Kamm (2010); Wiley VCH, available on-line in TUHH-library Powerpoint-Präsentations / selected Publications / further recommendations depending on the actual developments Industrial Biorefineries and White Biorefinery, by Pandey, Höfer, Larroche, Taherzadeh, Nampoothiri (Eds.); (2014 book development in progress)

Course L0974: Biorefinery Technologie	
Typ	Recitation Section (small)
Hrs/wk	1
CP	1
Workload in Hours	Independent Study Time 16, Study Time in Lecture 14
Lecturer	Dr. Ina Körner
Language	EN
Cycle	WiSe
Content	1.) Selection of a topic within the thematic area "Biorefinery Technologie" from a given list or self-selected. 2.) Self-dependent recherches to the topic. 3.) Preparation of a written elaboration. 4.) Presentation of the results in the group.
Literature	Vom Thema abhängig. Eigene Recherchen nötig. Depending on the topic. Own recheches necessary.

Course L0892: Bioresource Management	
Typ	Lecture
Hrs/wk	2
CP	2
Workload in Hours	Independent Study Time 32, Study Time in Lecture 28
Lecturer	Dr. Ina Körner
Language	EN
Cycle	WiSe
Content	In the context of limited fossil resources, climate change mitigation and increasing population growth, Bioresources has a special role. They have to feed the population and in the same time they are important for material production such as pulp and paper or construction materials. Moreover they become more and more important in chemical industry and in energy provision as fossil substitution. Although Bioresources are renewable, they are also considered as limited resources. The availability of land on our planet is the main limitation factor. The sustainable and reliable supply of non-food biomass feedstock is a critical for successful and long term perspective on production of bioenergy and other bio-based products. As the consequence, the increasing competition and shortages continue to happen at the traditional sectors. On the other side, huge unused but potentials residue on waste and wastewater sector exist. Nowadays, a lot of activities to develop better processes, to create new bio-based products in order to become more efficient, the inclusion of secondary and tertiary bio-resources in the valorisation chain are going on. The lecture deals with the current state-of-the-art of bioresource management. It shows deficits and potentials for improvement especially in the sector of utilization of organic residues for material and energy generation: <i>Lectures on:</i> • Bioresource generation and utilization including lost potentials today • Basic biological, mechanical, physico-chemical and logistical processes • The conflict of material vs. energy generation from wood / waste wood • The basics of pulp & paper production including waste paper recycling • The Pros and Cons from biogas and compost production <i>Special lectures by invited guests from research and practice:</i> • Pathways of waste organics on the example of Hamburg's City Cleaning Company • Utilization options of landscaping materials on the example of grass • Increase of process efficiency of anaerobic digestions • Decision support tools on the example of an municipality in Indonesia <i>Optional: Technical visits</i>
Literature	Power-Point presentations in STUD-IP

Course L0893: Bioresource Management	
Typ	Recitation Section (small)
Hrs/wk	1
CP	1
Workload in Hours	Independent Study Time 16, Study Time in Lecture 14
Lecturer	Dr. Ina Körner
Language	EN
Cycle	WiSe
Content	See interlocking course
Literature	See interlocking course

Module M2004: Sustainable Circular Economy				
Courses				
Title		Typ	Hrs/wk	CP
Circular Economy (L3264)		Seminar	2	3
Environment and Sustainability (L0319)		Lecture	2	3
Module Responsible	Prof. Kerstin Kuchta			
Admission Requirements	None			
Recommended Previous Knowledge	none			
Educational Objectives	After taking part successfully, students have reached the following learning results			
Professional Competence	Knowledge Students are able to describe single techniques and to give an overview for the field of safety and risk assessment, Circular Economy as well as environmental and sustainable engineering, in detail: - basics in safety and reliability of technical facilities - risk assessment and reliability analysis methods - Circularity of material - Identification and evaluation of material flows - energy production and supply - sustainable product design Skills Students are able apply interdisciplinary system-oriented methods for Circularity and risk assessment as well as sustainability reporting. They can evaluate the effort and costs for processes and select economically feasible treatment concepts. Personal Competence Social Competence Autonomy Students can gain knowledge of the subject area from given sources and transform it to new questions. Furthermore, they can define targets for new application or research-oriented duties in for risk management and sustainability concepts accordance with the potential social, economic and cultural impact.			
Workload in Hours	Independent Study Time 124, Study Time in Lecture 56			
Credit points	6			
Course achievement	None			
Examination	Written elaboration			
Examination duration and scale	Elaboration and presentation (45 minutes in groups)			
Assignment for the Following Curricula	Civil Engineering: Core Qualification: Compulsory Bioprocess Engineering: Specialisation C - Bioeconomic Process Engineering, Focus Management and Controlling: Elective Compulsory Chemical and Bioprocess Engineering: Specialisation General Process Engineering: Elective Compulsory Chemical and Bioprocess Engineering: Specialisation Bioprocess Engineering: Elective Compulsory Chemical and Bioprocess Engineering: Specialisation Chemical Process Engineering: Elective Compulsory Chemical and Bioprocess Engineering: Specialisation Chemical and Bioprocess Engineering: Elective Compulsory Chemical and Bioprocess Engineering: Specialisation Chemical and Bioprocess Engineering: Elective Compulsory Environmental Engineering: Specialisation Energy and Resources: Elective Compulsory Mechanical Engineering - Product Development and Production: Specialisation Product Development: Elective Compulsory Mechanical Engineering - Product Development and Production: Specialisation Production: Elective Compulsory Mechanical Engineering - Product Development and Production: Specialisation Materials: Elective Compulsory Product Development, Materials and Production: Specialisation Product Development: Elective Compulsory Product Development, Materials and Production: Specialisation Production: Elective Compulsory Product Development, Materials and Production: Specialisation Materials: Elective Compulsory Water and Environmental Engineering: Core Qualification: Compulsory			

Course L3264: Circular Economy	
Typ	Seminar
Hrs/wk	2
CP	3
Workload in Hours	Independent Study Time 62, Study Time in Lecture 28
Lecturer	Dr. Marco Ritzkowski
Language	EN
Cycle	WiSe
Content	The seminar deals with the basic idea as well as with core elements, advantages and challenges of the circular economy using concrete examples. The transition from linear to circular material flows is illustrated using the aspects of product design, reuse, recycling, avoidance (resource conservation) and the sharing economy. The concepts and examples presented are discussed with the students, deepened in group work and then presented.
Literature	Suitable literature will be announced in the course.

Course L0319: Environment and Sustainability	
Typ	Lecture
Hrs/wk	2
CP	3
Workload in Hours	Independent Study Time 62, Study Time in Lecture 28
Lecturer	Prof. Kerstin Kuchta
Language	EN
Cycle	WiSe
Content	This course presents actual methodologies and examples of environmental relevant, sustainable technologies, concepts and strategies in the field of energy supply, product design, water supply, waste water treatment or mobility. The following list shows examples: • Production and use of biochar • Energy production with algae • Environmentally friendly product design • Clean development mechanisms • Democracy and energy • Alternative mobility
Literature	Wird in der Veranstaltung bekannt gegeben.

Module M2006: Waste Treatment and Recycling				
Courses				
Title		Type	Hrs/wk	CP
Planning of waste treatment plants (L3267)		Project-/problem-based Learning	3	3
Recycling technologies and thermal waste treatment (L3265)		Lecture	2	2
Recycling technologies and thermal waste treatment (L3266)		Recitation Section (small)	1	1
Module Responsible	Prof. Kerstin Kuchta			
Admission Requirements	None			
Recommended Previous Knowledge	- Basics of thermo dynamics - Basics of fluid dynamics - fluid dynamics chemistry			
Educational Objectives	After taking part successfully, students have reached the following learning results			
Professional Competence	<i>Knowledge</i> The students can name, describe current issue and problems in the field of waste treatment (mechanical, chemical and thermal) and contemplate them in the context of their field. The industrial application of unit operations as part of process engineering is explained by actual examples of waste technologies . Compostion, particle sizes, transportation and dosing of wastes are described as important unit operations . Students will be able to design and design waste treatment technology equipment. <i>Skills</i> The students are able to select suitable processes for the treatment of wastes or raw material with respect to their characteristics and the process aims. They can evaluate the efforts and costs for processes and select economically feasible treatment concepts. Personal Competence <i>Social Competence</i> Students can - respectfully work together as a team and discuss technical tasks - participate in subject-specific and interdisciplinary discussions, - develop cooperated solutions - promote the scientific development and accept professional constructive criticism. <i>Autonomy</i> Students can independently tap knowledge of the subject area and transform it to new questions. They are capable, in consultation with supervisors, to assess their learning level and define further steps on this basis. Furthermore, they can define targets for new application-or research-oriented duties in accordance with the potential social, economic and cultural impact.			
Workload in Hours				
Credit points				
Course achievement				
Examination				
Examination duration and scale	120 min			
Assignment for the Following Curricula	Civil Engineering: Specialisation Water and Traffic: Elective Compulsory Bioprocess Engineering: Specialisation A - General Bioprocess Engineering: Elective Compulsory Chemical and Bioprocess Engineering: Specialisation General Process Engineering: Elective Compulsory Chemical and Bioprocess Engineering: Specialisation Bioprocess Engineering: Elective Compulsory Chemical and Bioprocess Engineering: Specialisation Chemical Process Engineering: Elective Compulsory Chemical and Bioprocess Engineering: Specialisation Chemical and Bioprocess Engineering: Elective Compulsory Chemical and Bioprocess Engineering: Specialisation Chemical and Bioprocess Engineering: Elective Compulsory Environmental Engineering: Specialisation Energy and Resources: Elective Compulsory International Management and Engineering: Specialisation II. Renewable Energy: Elective Compulsory Renewable Energies: Specialisation Bioenergy Systems: Elective Compulsory Process Engineering: Specialisation Chemical Process Engineering: Elective Compulsory Process Engineering: Specialisation Process Engineering: Elective Compulsory Process Engineering: Specialisation Environmental Process Engineering: Elective Compulsory Water and Environmental Engineering: Specialisation Environment: Compulsory Water and Environmental Engineering: Specialisation Cities: Elective Compulsory			

Course L3267: Planning of waste treatment plants	
Typ	Project-/problem-based Learning
Hrs/wk	3
CP	3
Workload in Hours	Independent Study Time 48, Study Time in Lecture 42
Lecturer	Prof. Rüdiger Siechau
Language	EN
Cycle	WiSe
Content	The focus is on getting to know the organization and practice of waste management companies. Topics such as planning, financing and logistics will be discussed and there will be an excursion (waste incineration plant, vehicle fleet and collection systems / containers). Project based learning: You will be given a task to work on independently in groups of 4 to 6 students. All tools and data needed for the project work will be discussed in the lecture "Recycling Technologies and Thermal Waste Treatment". Course documents can be downloaded from StudIP. Communication during the project work also takes place via StudIP.
Literature	- Einführung in die Abfallwirtschaft; Martin Kranert, Klaus Cord-Landwehr (Hrsg.); Vieweg + Teubner Verlag; 2010 - PowerPoint Präsentationen in Stud IP

Course L3265: Recycling technologies and thermal waste treatment	
Typ	Lecture
Hrs/wk	2
CP	2
Workload in Hours	Independent Study Time 32, Study Time in Lecture 28
Lecturer	Prof. Kerstin Kuchta
Language	EN
Cycle	WiSe
Content	- Introduction, actual state-of-the-art of waste incineration, aims. legal background, reaction principals - basics of incineration processes: waste composition, calorific value, calculation of air demand and flue gas composition - Incineration techniques: grate firing, ash transfer, boiler - Flue gas cleaning: Volume, composition, legal frame work and emission limits, dry treatment, scrubber, de-nox techniques, dioxin elimination, Mercury elimination - Ash treatment: Mass, quality, treatment concepts, recycling, disposal
Literature	Thomé-Kozmiensky, K. J. (Hrsg.): Thermische Abfallbehandlung Bande 1-7. EF-Verlag für Energie- und Umwelttechnik, Berlin, 196 - 2013.

Course L3266: Recycling technologies and thermal waste treatment	
Typ	Recitation Section (small)
Hrs/wk	1
CP	1
Workload in Hours	Independent Study Time 16, Study Time in Lecture 14
Lecturer	Prof. Kerstin Kuchta
Language	EN
Cycle	WiSe
Content	See interlocking course
Literature	See interlocking course

Specialization Environment and Climate

Module M1724: Smart Monitoring

Courses

Title	Typ	Hrs/wk	CP
Smart Monitoring (L2762)	Integrated Lecture	2	2
Smart Monitoring (L2763)	Recitation Section (small)	2	4
Module Responsible	Prof. Kay Smarsly		
Admission Requirements	None		
Recommended Previous Knowledge	Basic knowledge or interest in object-oriented modeling, programming, and sensor technologies are helpful. Interest in modern research and teaching areas, such as Internet of Things, Industry 4.0 and cyber-physical systems, as well as the will to deepen skills of scientific working, are required. Basic knowledge in scientific writing and good English skills.		
Educational Objectives	After taking part successfully, students have reached the following learning results		
Professional Competence	The students will become familiar with the principles and practices of smart monitoring. The students will be able to design decentralized smart systems to be applied for continuous (remote) monitoring of systems in the built and in the natural environment. In addition, the students will learn to design and to implement intelligent sensor systems using state-of-the-art data analysis techniques, modern software design concepts, and embedded computing methodologies. Besides lectures, project work is also part of this module, which will be conducted throughout the semester and will contribute to the grade. In small groups, the students will design smart monitoring systems that integrate a number of "intelligent" sensors to be implemented by the students. Specific focus will be put on the application of machine learning techniques. The smart monitoring systems will be mounted on real-world (built or natural) systems, such as bridges or slopes, or on scaled lab structures for validation purposes. The outcome of every group will be documented in a paper. All students of this module will "automatically" participate with their smart monitoring system in the annual "Smart Monitoring" competition. The written papers and oral examinations form the final grades. The module will be taught in English. Limited enrollment. The students will gain insights into operating state-of-the-art smart sensor systems, used for monitoring a wide range of physical processes relevant to engineering, such as environmental, structural, or comfort monitoring. The students will be capable of devising monitoring strategies of physical processes as part of group projects, tailored to their knowledge backgrounds, and to implement the strategies in smart wireless sensor nodes, using embedded computing and programming. Finally, the students will be able to document the findings of their projects in short reports. The students will be able to work in groups, share parts of the work for their projects, and develop communication skills, towards achieving the common project goals. The students will be able to gain a solid basis on approaching and solving problems in engineering, as well as on documenting results, through their involvement in their monitoring group projects.		
<i>Knowledge</i>			
<i>Skills</i>			
Personal Competence			
<i>Social Competence</i>			
<i>Autonomy</i>			
Workload in Hours	Independent Study Time 124, Study Time in Lecture 56		
Credit points	6		
Course achievement	None		
Examination	Written elaboration		
Examination duration and scale	10 pages of work with 15-minute oral presentation		
Assignment for the Following Curricula	Civil Engineering: Specialisation Water and Traffic: Elective Compulsory Civil Engineering: Specialisation Geotechnical Engineering: Elective Compulsory Civil Engineering: Specialisation Coastal Engineering: Elective Compulsory Civil Engineering: Specialisation Structural Engineering: Elective Compulsory Computer Science: Specialisation II: Intelligence Engineering: Elective Compulsory Environmental Engineering: Specialisation Energy and Resources: Elective Compulsory Environmental Engineering: Specialisation Environment and Climate: Elective Compulsory Environmental Engineering: Specialisation Water Quality and Water Engineering: Elective Compulsory Mechatronics: Technical Complementary Course: Elective Compulsory Mechatronics: Core Qualification: Elective Compulsory Theoretical Mechanical Engineering: Specialisation Robotics and Computer Science: Elective Compulsory Water and Environmental Engineering: Specialisation Cities: Elective Compulsory Water and Environmental Engineering: Specialisation Environment: Elective Compulsory Water and Environmental Engineering: Specialisation Water: Elective Compulsory		

Course L2762: Smart Monitoring	
Typ	Integrated Lecture
Hrs/wk	2
CP	2
Workload in Hours	Independent Study Time 32, Study Time in Lecture 28
Lecturer	Prof. Kay Smarsly
Language	EN
Cycle	SoSe
Content	In this course, principles of smart monitoring will be taught, focusing on modern concepts of data acquisition, data storage, and data analysis. Also, fundamentals of intelligent sensors and embedded computing will be illuminated. Autonomous software and decentralized data processing are further crucial parts of the course, including concepts of the Internet of Things, Industry 4.0 and cyber-physical systems. Furthermore, measuring principles, data acquisition systems, data management and data analysis algorithms will be discussed. Besides the theoretical background, numerous practical examples will be shown to demonstrate how smart monitoring may advantageously be used for assessing the condition of systems in the built or natural environment.
Literature	The course contents couples different fields, such as signal processing, sensing technologies, data analytics, environmental engineering, civil engineering, artificial intelligence, database systems, and many more. The basics will be taught in this course. However, specific literature that covers all these topics does not exist. Instead, literature will be referenced in the lectures, all of which are papers that are freely available online.

Course L2763: Smart Monitoring	
Typ	Recitation Section (small)
Hrs/wk	2
CP	4
Workload in Hours	Independent Study Time 92, Study Time in Lecture 28
Lecturer	Prof. Kay Smarsly
Language	EN
Cycle	SoSe
Content	The contents of the exercises are based on the lecture contents. In addition to the exercises, project work will be conducted throughout the semester, which will consume the majority of the workload. As part of the project work, students will design smart monitoring systems that will be tested in the laboratory or in the field. As mentioned in the module description, the students will participate in the "Smart Monitoring" competition, hosted annually by the Institute of Digital and Autonomous Construction. Students are encouraged to contribute their own ideas. The tools required to implement the smart monitoring systems will be taught in the group exercises as well as through external sources, such as video tutorials and literature.
Literature	The course contents couples different fields, such as signal processing, sensing technologies, data analytics, environmental engineering, civil engineering, artificial intelligence, database systems, and many more. The basics will be taught in this course. However, specific literature that covers all these topics does not exist. Instead, literature will be referenced in the lectures, all of which are papers that are freely available online.

Module M0858: Coastal Hydraulic Engineering I			
Courses			
Title	Typ	Hrs/wk	CP
Basics of Coastal Engineering (L0807)	Lecture	3	4
Basics of Coastal Engineering (L1413)	Project-/problem-based Learning	1	2
Module Responsible	Prof. Peter Fröhle		
Admission Requirements	None		
Recommended Previous Knowledge	Basics of hydraulic engineering, hydrology and hydromechanics		
Educational Objectives	After taking part successfully, students have reached the following learning results		
Professional Competence	<i>Knowledge</i> The students are able to define and explain the basic concepts of coastal engineering and port engineering. They are able to apply the concepts to selected practical problems of coastal engineering. Students can define and determine the basics for design and dimensioning of coastal engineering constructions. <i>Skills</i> The students are capable to apply basic design approaches to selected and pre-defined design tasks in coastal engineering. <i>Social Competence</i> The students are able to deploy their gained knowledge in applied problems such as the design of coastal protection structures. Additionally, they will be able to work in team with engineers of other disciplines, for instance designing of coastal breakwaters. <i>Autonomy</i> The students will be able to independently extend their knowledge and apply it to new problems.		
Personal Competence			
Workload in Hours			
Credit points			
Course achievement	None		
Examination	Written exam		
Examination duration and scale	The duration of the examination is 2 hours. The examination includes tasks with respect to the general understanding of the lecture contents and calculations tasks.		
Assignment for the Following Curricula	Civil Engineering: Specialisation Coastal Engineering: Compulsory Civil Engineering: Specialisation Geotechnical Engineering: Compulsory Civil Engineering: Specialisation Structural Engineering: Elective Compulsory Environmental Engineering: Specialisation Environment and Climate: Elective Compulsory Environmental Engineering: Specialisation Water Quality and Water Engineering: Elective Compulsory International Management and Engineering: Specialisation II. Civil Engineering: Elective Compulsory Water and Environmental Engineering: Specialisation Environment: Elective Compulsory Water and Environmental Engineering: Specialisation Water: Elective Compulsory		

Course L0807: Basics of Coastal Engineering	
Typ	Lecture
Hrs/wk	3
CP	4
Workload in Hours	Independent Study Time 78, Study Time in Lecture 42
Lecturer	Prof. Peter Fröhle
Language	EN
Cycle	SoSe
Content	- Basics of planning and design - Water levels - Currents - Waves - Ice - Planning and Design in Coastal Engineering - Functional and constructional design - Determination of design parameters - Design-approaches - Filter - Rubble mound constructions - Piles - Vertical constructions
Literature	Coastal Engineering Manual, CEM Vorlesungsumdruck

Course L1413: Basics of Coastal Engineering	
Typ	Project-/problem-based Learning
Hrs/wk	1
CP	2
Workload in Hours	Independent Study Time 46, Study Time in Lecture 14
Lecturer	Prof. Peter Fröhle
Language	EN
Cycle	SoSe
Content	See interlocking course
Literature	See interlocking course

Module M1721: Water and Environment: Theory and Application				
Courses				
Title		Typ	Hrs/wk	CP
Water and Environment (L2754)		Project-/problem-based Learning	3	3
Water and Environment (L2753)		Lecture	3	3
Module Responsible	Prof. Nima Shokri			
Admission Requirements	None			
Recommended Previous Knowledge	Basic knowledge in water and environmental research, Hydrology			
Educational Objectives	After taking part successfully, students have reached the following learning results			
Professional Competence	Common research tools and techniques together with the fundamental knowledge relevant to multi-scale and multi-phase challenges present in water and environmental research will be discussed in this module. Both theory and application will be considered. In addition to the fundamental knowledge, the students will be exposed to several analytical, experimental and numerical tools and techniques relevant to water and environmental research at different scales. This will provide the students with an excellent opportunity to improve their skills on multiple fronts which will be useful in their future career. Developing teamwork and problem solving skills through Research-Based Teaching approaches will be at the core of this module. The students will be involved in writing individual reports and presentation. This will contribute to the students' ability and willingness to work independently and responsibly.			
Knowledge				
Skills				
Personal Competence				
Social Competence				
Autonomy				
Workload in Hours	Independent Study Time 96, Study Time in Lecture 84			
Credit points	6			
Course achievement	None			
Examination	Subject theoretical and practical work			
Examination duration and scale	Report and Presentation			
Assignment for the Following Curricula	Civil Engineering: Specialisation Coastal Engineering: Elective Compulsory Civil Engineering: Specialisation Water and Traffic: Elective Compulsory Environmental Engineering: Specialisation Environment and Climate: Elective Compulsory Water and Environmental Engineering: Specialisation Cities: Elective Compulsory Water and Environmental Engineering: Specialisation Water: Elective Compulsory Water and Environmental Engineering: Specialisation Environment: Compulsory			

Course L2754: Water and Environment	
Typ	Project-/problem-based Learning
Hrs/wk	3
CP	3
Workload in Hours	Independent Study Time 48, Study Time in Lecture 42
Lecturer	Dr. Salome Shokri-Kuehni
Language	EN
Cycle	SoSe
Content	See interlocking course
Literature	See interlocking course

Course L2753: Water and Environment	
Typ	Lecture
Hrs/wk	3
CP	3
Workload in Hours	Independent Study Time 48, Study Time in Lecture 42
Lecturer	Prof. Nima Shokri
Language	EN
Cycle	SoSe
Content	Research based learning: The students will be engaged in active research focused on water and environmental related challenges. The required knowledge and tools will be discussed during the semester.
Literature	NA

Module M1980: Field measurements for environmental studies				
Courses				
Title		Typ	Hrs/wk	CP
Field measurements for environmental studies: Application (L3231)		Project-/problem-based Learning	3	4
Field measurements for environmental studies: Theory (L3230)		Lecture	1	2
Module Responsible	Prof. Nima Shokri			
Admission Requirements	None			
Recommended Previous Knowledge	Basic Mathematics, Physics, Hydrology			
Educational Objectives	After taking part successfully, students have reached the following learning results			
Professional Competence				
<i>Knowledge</i>	Upon completing this module, students will understand the mechanisms controlling water flow and transport processes in soil. They will also be capable of using sensors and devices to measure soil conditions and meteorological parameters.			
<i>Skills</i>	Field-scale experiments are planned, and students will learn to use various environmental measurement tools and analyze the collected data in this module.			
Personal Competence				
<i>Social Competence</i>	The students will work together in groups to conduct the measurements and discuss the findings and results.			
<i>Autonomy</i>	Preparing the report of the practical activities will help the students to strengthen their skills for writing scientific reports, data analysis and interpretations. This will be useful for the students' master thesis and documentation in general.			
Workload in Hours	Independent Study Time 124, Study Time in Lecture 56			
Credit points	6			
Course achievement	None			
Examination	Subject theoretical and practical work			
Examination duration and scale	Report & Präsentation			
Assignment for the Following Curricula	Environmental Engineering: Specialisation Environment and Climate: Elective Compulsory Water and Environmental Engineering: Specialisation Environment: Elective Compulsory			

Course L3231: Field measurements for environmental studies: Application	
Typ	Project-/problem-based Learning
Hrs/wk	3
CP	4
Workload in Hours	Independent Study Time 78, Study Time in Lecture 42
Lecturer	Dr. Milad Aminzadeh
Language	EN
Cycle	SoSe
Content	Students will be trained to use state-of-the-art environmental measurement tools and sensors, including tensiometers, penetrometers, soil heat characterization instruments, electrical resistivity tomography equipment, and land surface energy balance components. This hands-on training will take place in the Living Lab on Soil Health, Water, and Climate, operated by the Institute of Geo-Hydroinformatics at TUHH. The dates for the full day field activities will be planned at the beginning of the semester with the students.
Literature	- Environmental Soil Physics, by Daniel Hillel - Soil Physics, Sixth Edition, by William A. Jury and Robert Horton

Course L3230: Field measurements for environmental studies: Theory	
Typ	Lecture
Hrs/wk	1
CP	2
Workload in Hours	Independent Study Time 46, Study Time in Lecture 14
Lecturer	Dr. Milad Aminzadeh
Language	EN
Cycle	SoSe
Content	The students will learn the fundamentals of environmental sensing technology that applies to soil health and processes. They will develop the ability to work with governing equations that describe transport processes in soil. Additionally, they will gain experience in understanding and interpreting data collected in the Living Lab.
Literature	- Environmental Soil Physics, by Daniel Hillel - Soil Physics, Sixth Edition, by William A. Jury and Robert Horton

Module M1900: Study work Environment and Climate				
Courses				
Title	Typ		Hrs/wk	CP
Module Responsible	Dozenten des Studiengangs			
Admission Requirements	None			
Recommended Previous Knowledge				
Educational Objectives	After taking part successfully, students have reached the following learning results			
Professional Competence <i>Knowledge</i>	Students are able to demonstrate their detailed knowledge in an area of environmental engineering. The students are qualified to project climate and environmental protection-oriented projects and to independently define research tasks for the theoretical and experimental investigation of environmental problems. They are able to give examples of the state of development and application and discuss these critically, taking into account current problems and framework conditions in science and society. The students are able to independently define a solution strategy for a basic, application-oriented or practical problem from the field of environmental engineering and to outline individual solution approaches. In doing so, they are able to proceed in a theory-oriented manner and include current safety, ecological, ethical and economic aspects according to the state of the art in science and related social discussions. They can use the scientific working techniques they have chosen for their own project work, they can present them in detail and critically discuss them.			
<i>Skills</i>	Students are able to independently select methodological approaches for project work and justify this selection in terms of content. They can explain how they relate approaches or methods to the specific field of application in a solution-oriented manner and adapt them to the application context. They can outline the main points and further developments that go beyond the project.			
Personal Competence <i>Social Competence</i>	Students are able to prepare the relevance and cut of their project task, the work steps and sub-problems for discussion and debate in larger groups, guide the discussions and give feedback to colleagues on their projects.			
<i>Autonomy</i>	The students are able to independently plan and document the work steps and processes necessary to complete the coursework, taking into account specified deadlines. This includes being able to obtain current scientific information in a goal-oriented manner. Furthermore, they are able to obtain feedback on the progress of work from experts in the field in order to achieve high-quality work results based on the state of the art in science and technology.			
Workload in Hours	Independent Study Time 360, Study Time in Lecture 0			
Credit points	12			
Course achievement	None			
Examination	Study work			
Examination duration and scale	depending on task			
Assignment for the Following Curricula	Environmental Engineering: Specialisation Environment and Climate: Compulsory			

Module M0859: Coastal Hydraulic Engineering II			
Courses			
Title	Typ	Hrs/wk	CP
Coastal- and Flood Protection (L0808)	Lecture	2	3
Coastal- and Flood Protection (L1415)	Project-/problem-based Learning	1	1
Maintenance and Defence of Flood Protection Structures (L1411)	Lecture	2	2
Module Responsible	Prof. Peter Fröhle		
Admission Requirements	None		
Recommended Previous Knowledge	Coastal Engineering I		
Educational Objectives	After taking part successfully, students have reached the following learning results		
Professional Competence	Knowledge The students have the capability to define and explain in detail the important aspects of erosion protection and flood protection and are able to apply the aspects to practical coastal protection problems. They are able to design and dimension important coastal protection measures from the functional and from the constructional point of view. Skills The students are able to select design approaches for the functional and constructional design of erosion and flood protection measures and apply these approaches to practical design tasks. Personal Competence Social Competence The students are able to deploy their gained knowledge in applied problems such as the functional and constructive design of coastal and flood protection structures. Additionally, they will be able to work in team with engineers of other disciplines. Autonomy The students will be able to independently extend their knowledge and apply it to new problems.		
Workload in Hours	Independent Study Time 110, Study Time in Lecture 70		
Credit points	6		
Course achievement	None		
Examination	Written exam		
Examination duration and scale	The duration of the examination is 130 min. The examination includes tasks with respect to the general understanding of the lecture contents and calculations tasks.		
Assignment for the Following Curricula	Civil Engineering: Specialisation Coastal Engineering: Compulsory Civil Engineering: Specialisation Geotechnical Engineering: Elective Compulsory Civil Engineering: Specialisation Structural Engineering: Elective Compulsory Environmental Engineering: Specialisation Environment and Climate: Elective Compulsory Environmental Engineering: Specialisation Water Quality and Water Engineering: Elective Compulsory Water and Environmental Engineering: Specialisation Environment: Elective Compulsory Water and Environmental Engineering: Specialisation Water: Elective Compulsory		

Course L0808: Coastal- and Flood Protection	
Typ	Lecture
Hrs/wk	2
CP	3
Workload in Hours	Independent Study Time 62, Study Time in Lecture 28
Lecturer	Prof. Peter Fröhle
Language	EN
Cycle	WiSe
Content	Protection of sandy coasts • Sediment transport • Morphology • Technical solution for the protection of sandy coasts ◦ Construction in direction of the coast ◦ Constructions perpendicular to the coast ◦ Other Concept • Calculation approaches and numerical models Flood Protection • Classification of constructions / measures • Dikes • Dunes • Foreland - constructions • Flood-Protection Walls • Drainage of the hinterland
Literature	Vorlesungsumdruck Coastal Engineering Manual CEM

Course L1415: Coastal- and Flood Protection	
Typ	Project-/problem-based Learning
Hrs/wk	1
CP	1
Workload in Hours	Independent Study Time 16, Study Time in Lecture 14
Lecturer	Prof. Peter Fröhle
Language	EN
Cycle	WiSe
Content	See interlocking course
Literature	See interlocking course

Course L1411: Maintenance and Defence of Flood Protection Structures	
Typ	Lecture
Hrs/wk	2
CP	2
Workload in Hours	Independent Study Time 32, Study Time in Lecture 28
Lecturer	Dr. Olaf Müller
Language	EN
Cycle	WiSe
Content	• Dike protection • Maintenance of flood protection measures
Literature	Vorlesungsumdruck

Module M1720: Emerging Trends in Environmental Engineering			
Courses			
Title	Type	Hrs/wk	CP
Environmental Research Trends (L2752)	Seminar	2	2
Microplastics in Environment (L2750)	Lecture	2	2
Scientific Communication and Methods (L2751)	Lecture	1	2
Module Responsible	Prof. Nima Shokri		
Admission Requirements	None		
Recommended Previous Knowledge	Basic knowledge on water, soil and environmental research.		
Educational Objectives	After taking part successfully, students have reached the following learning results		
Professional Competence	<i>Knowledge</i> The students will be exposed to up-to-date research topics focused on soil, water and climate related challenges with a particular focus on the effects of microplastics in environment. Data analysis, data measurement, curation and presentation will be other skills that the students will develop in this module. <i>Skills</i> Students' research skills will be improved in this module. How to prepare and deliver an effective presentation, how to write an abstract, research paper and proposal will be discussed in this module. Moreover, through Research-Based Learning approaches, the students will be exposed to current research trends in environmental engineering. Personal Competence <i>Social Competence</i> Developing teamwork and problem solving skills through Research-Based Teaching approaches will be at the core of this module. <i>Autonomy</i> The students will be involved in writing individual reports and presentation. This will contribute to the students' ability and willingness to work independently and responsibly.		
Workload in Hours	Independent Study Time 110, Study Time in Lecture 70		
Credit points	6		
Course achievement	None		
Examination	Subject theoretical and practical work		
Examination duration and scale	Report and Presentation		
Assignment for the Following Curricula	Civil Engineering: Specialisation Water and Traffic: Elective Compulsory Environmental Engineering: Specialisation Environment and Climate: Elective Compulsory Water and Environmental Engineering: Specialisation Cities: Elective Compulsory Water and Environmental Engineering: Specialisation Environment: Elective Compulsory Water and Environmental Engineering: Specialisation Water: Elective Compulsory		

Course L2752: Environmental Research Trends	
Typ	Seminar
Hrs/wk	2
CP	2
Workload in Hours	Independent Study Time 32, Study Time in Lecture 28
Lecturer	Dr. Salome Shokri-Kuehni
Language	EN
Cycle	WiSe
Content	Introduction - course objectives, expectations and format Analyzing the Audience, purpose and occasion Constructing and delivering effective technical presentations How to write an abstract How to write a scientific paper Developing competitive and persuasive research proposals Databases and resources available for water and environmental research Individual proposal on water and environmental research Individual project on water and environmental research Presentation on water and environmental research
Literature	- The Craft of Scientific Writing Fourth edition Author: Michael Alley Springer-Verlag New York, Copyright 2018, DOI 10.1007/978-1-4419-8288-9 - Supplemental materials and web links which will be available to registered students.

Course L2750: Microplastics in Environment	
Typ	Lecture
Hrs/wk	2
CP	2
Workload in Hours	Independent Study Time 32, Study Time in Lecture 28
Lecturer	Prof. Nima Shokri
Language	EN
Cycle	WiSe
Content	- Introduction, objectives, expectations, format, importance - Sources of microplastics in environment - Microplastics sampling; Characterization of microplastics - Distribution of microplastics in terrestrial environments - Fate of microplastics in terrestrial environments - Project discussion - Effects of microplastics on terrestrial environments - Health risks of microplastics in environments - Project presentations by all students
Literature	- Microplastics in Terrestrial Environments (2021), Edited by Defu He and Yongming Luo - Particulate Plastics in Terrestrial and Aquatic Environments (2020), Edited by Nanthi S. Bolan et al. - Microplastic Pollutants (2017), by Christopher B. Crawford and Brian Quinn

Course L2751: Scientific Communication and Methods	
Typ	Lecture
Hrs/wk	1
CP	2
Workload in Hours	Independent Study Time 46, Study Time in Lecture 14
Lecturer	Prof. Nima Shokri
Language	EN
Cycle	WiSe
Content	Introduction - course objectives, expectations and format Analyzing the Audience, purpose and occasion Constructing and delivering effective technical presentations How to write an abstract How to create a scientific poster How to write a scientific paper Developing competitive and persuasive research proposals Individual project (report and presentation) related to soil, water and environmental research
Literature	• The Craft of Scientific Writing Fourth edition Author: Michael Alley Springer-Verlag New York, Copyright 2018, DOI 10.1007/978-1-4419-8288-9 • Supplemental materials and web links which will be available to registered students.

Module M1779: Sustainable Nature-based Coastal Protection in a Changing Climate (SeaPiaC)				
Courses				
Title	Typ		Hrs/wk	CP
Sustainable Nature-based Coastal Protection in a Changing Climate (SeaPiaC) (L2926)	Project-/problem-based Learning		4	6
Module Responsible	Prof. Peter Fröhle			
Admission Requirements	None			
Recommended Previous Knowledge	- Hydraulic Engineering - Hydromechanics, Hydraulics - Fundamentals of Coastal Engineering, Coastal- and Flood Protection			
Educational Objectives	After taking part successfully, students have reached the following learning results			
Professional Competence <i>Knowledge</i> - Climate and Climate Change - General Impacts of Climate Change on Wind Regime and Water Cycle - Consequences of Climate Change for Coastal Processes - Coastal Protection in Taiwan and Germany - Fundamentals of Climate Adaptation - Nature-based Solutions (NBS) for Coastal Protection <i>Skills</i> - Critical thinking: analysis of processes and relations, assessment of needs for action - Creative thinking: development of adaptation strategies and adaptation measures - Practical thinking: inclusion of restrictions, application of calculation approaches, methods, numerical models, planning methods - Consideration of complex tasks Personal Competence <i>Social Competence</i> - Working in heterogenous groups - Working in international groups - Working with different scientific / non-scientific disciplines - Self reflection <i>Autonomy</i> - Application oriented use of knowledge and skills - Autonomous work on complex tasks				
Workload in Hours	Independent Study Time 124, Study Time in Lecture 56			
Credit points	6			
Course achievement	None			
Examination	Written elaboration			
Examination duration and scale	Preparation of a written report on a complex task with a presentation and subsequent discussion. The work on the complex task happens in the course of the lecture.			
Assignment for the Following Curricula	Civil Engineering: Specialisation Coastal Engineering: Elective Compulsory Civil Engineering: Specialisation Geotechnical Engineering: Elective Compulsory Civil Engineering: Specialisation Structural Engineering: Elective Compulsory Civil Engineering: Specialisation Water and Traffic: Elective Compulsory Environmental Engineering: Specialisation Environment and Climate: Elective Compulsory Water and Environmental Engineering: Specialisation Cities: Elective Compulsory Water and Environmental Engineering: Specialisation Environment: Elective Compulsory Water and Environmental Engineering: Specialisation Water: Elective Compulsory			

Course L2926: Sustainable Nature-based Coastal Protection in a Changing Climate (SeaPiaC)	
Typ	Project-/problem-based Learning
Hrs/wk	4
CP	6
Workload in Hours	Independent Study Time 124, Study Time in Lecture 56
Lecturer	Prof. Peter Fröhle
Language	EN
Cycle	WiSe
Content	• Climate and Climate Change • General Impacts of Climate Change on Wind Regime and Water Cycle • Consequences of Climate Change for Coastal Processes • Coastal Protection in Taiwan and Germany • Fundamentals of Climate Adaptation • Nature-Based Solutions (NBS) for Coastal Protection
Literature	• Materials provided on eLearning Platform (HOOU Platform) • Depending on the main topics of the course in the respective year, the literature (recent papers) will be provided in the course-material or via StudIP.

Specialization Water Quality and Water Engineering

Module M0874: Wastewater Systems

Courses

Title	Typ	Hrs/wk	CP
Biological Wastewater Treatment (L0517)	Lecture	2	2
Biological Wastewater Treatment (L3122)	Recitation Section (large)	1	1
Advanced Wastewater Treatment (L0357)	Lecture	2	2
Advanced Wastewater Treatment (L0358)	Recitation Section (large)	1	1
Module Responsible	Dr. Joachim Behrendt		
Admission Requirements	None		
Recommended Previous Knowledge	Knowledge of wastewater management and the key processes involved in wastewater treatment.		
Educational Objectives	After taking part successfully, students have reached the following learning results		
Professional Competence	<i>Knowledge</i> Students are able to outline key areas of the full range of treatment systems in waste water management, as well as their mutual dependence for sustainable water protection. They can describe relevant economic, environmental and social factors. <i>Skills</i> Students are able to pre-design and explain the available wastewater treatment processes and the scope of their application in municipal and for some industrial treatment plants. <i>Social Competence</i> Social skills are not targeted in this module. <i>Autonomy</i> Students are in a position to work on a subject and to organize their work flow independently. They can also present on this subject.		
Workload in Hours	Independent Study Time 96, Study Time in Lecture 84		
Credit points	6		
Course achievement	None		
Examination	Written exam		
Examination duration and scale	120 min		
Assignment for the Following Curricula	Civil Engineering: Specialisation Structural Engineering: Elective Compulsory Civil Engineering: Specialisation Geotechnical Engineering: Elective Compulsory Civil Engineering: Specialisation Coastal Engineering: Elective Compulsory Civil Engineering: Specialisation Water and Traffic: Compulsory Bioprocess Engineering: Specialisation A - General Bioprocess Engineering: Elective Compulsory Environmental Engineering: Specialisation Water Quality and Water Engineering: Elective Compulsory International Management and Engineering: Specialisation II. Process Engineering and Biotechnology: Elective Compulsory International Management and Engineering: Specialisation II. Energy and Environmental Engineering: Elective Compulsory Process Engineering: Specialisation Environmental Process Engineering: Elective Compulsory Process Engineering: Specialisation Process Engineering: Elective Compulsory Water and Environmental Engineering: Specialisation Water: Compulsory Water and Environmental Engineering: Specialisation Environment: Elective Compulsory Water and Environmental Engineering: Specialisation Cities: Compulsory		

Course L0517: Biological Wastewater Treatment

Typ	Lecture
Hrs/wk	2
CP	2
Workload in Hours	Independent Study Time 32, Study Time in Lecture 28
Lecturer	Dr. Joachim Behrendt
Language	DE/EN
Cycle	SoSe
Content	Charaterisation of Wastewater Metobolism of Microorganisms Kinetic of mirobiotic processes Calculation of bioreactor for wastewater treatment Concepts of Wastewater treatment Design of WWTP Excursion to a WWTP Biofilms Biofim Reactors Anaerobic Wastewater and sledge treatment resources oriented sanitation technology Future challenges of wastewater treatment

Literature	Gujer, Willi Siedlungswasserwirtschaft : mit 84 Tabellen ISBN: 3540343296 (Gb.) URL: http://www.gbv.de/dms/bs/toc/516261924.pdf URL: http://deposit.d-nb.de/cgi-bin/dokserv?id=2842122&prov=M&dok_var=1&dok_ext=htm Berlin [u.a.] : Springer, 2007 TUB_HH_Katalog Henze, Mogens Wastewater treatment : biological and chemical processes ISBN: 3540422285 (Pp.) Berlin [u.a.] : Springer, 2002 TUB_HH_Katalog Imhoff, Karl (Imhoff, Klaus R.) Taschenbuch der Stadtentwässerung : mit 10 Tafeln ISBN: 3486263331 ((Gb.)) München [u.a.] : Oldenbourg, 1999 TUB_HH_Katalog Lange, Jörg (Otterpohl, Ralf; Steger-Hartmann, Thomas;) Abwasser : Handbuch zu einer zukunftsfähigen Wasserwirtschaft ISBN: 3980350215 (kart.) URL: http://www.gbv.de/du/services/agi/52567E5D44DA0809C12570220050BF25/000000700334 Donaueschingen-Pföhrn : Mall-Beton-Verl., 2000 TUB_HH_Katalog Mudrack, Klaus (Kunst, Sabine;) Biologie der Abwasserreinigung : 18 Tabellen ISBN: 382741427X URL: http://www.gbv.de/du/services/agi/94B581161B6EC747C1256E3F005A8143/420000114903 Heidelberg [u.a.] : Spektrum, Akad. Verl., 2003 TUB_HH_Katalog Tchobanoglous, George (Metcalf & Eddy, Inc., ;) Wastewater engineering : treatment and reuse ISBN: 0070418780 (alk. paper) ISBN: 0071122508 (ISE (*pbk)) Boston [u.a.] : McGraw-Hill, 2003 TUB_HH_Katalog Henze, Mogens Activated sludge models ASM1, ASM2, ASM2d and ASM3 ISBN: 1900222248 London : IWA Publ., 2002 TUB_HH_Katalog Kunz, Peter Umwelt-Bioverfahrenstechnik Vieweg, 1992 Bauhaus-Universität., Arbeitsgruppe Weiterbildendes Studium Wasser und Umwelt (Deutsche Vereinigung für Wasserwirtschaft, Abwasser und Abfall, ;) Abwasserbehandlung : Gewässerbelastung, Bemessungsgrundlagen, Mechanische Verfahren, Biologische Verfahren, Reststoffe aus der Abwasserbehandlung, Kleinkläranlagen ISBN: 3860682725 URL: http://www.gbv.de/dms/weimar/toc/513989765_toc.pdf URL: http://www.gbv.de/dms/weimar/abs/513989765_abs.pdf Weimar : Universitätsverl, 2006 TUB_HH_Katalog Deutsche Vereinigung für Wasserwirtschaft, Abwasser und Abfall DWA-Regelwerk Hennef : DWA, 2004 TUB_HH_Katalog Wiesmann, Udo (Choi, In Su; Dombrowski, Eva-Maria;) Fundamentals of biological wastewater treatment ISBN: 3527312196 (Gb.) URL: http://deposit.ddb.de/cgi-bin/dokserv?id=2774611&prov=M&dok_var=1&dok_ext=htm Weinheim : WILEY-VCH, 2007 TUB_HH_Katalog
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Course L3122: Biological Wastewater Treatment	
Typ	Recitation Section (large)
Hrs/wk	1
CP	1
Workload in Hours	Independent Study Time 16, Study Time in Lecture 14
Lecturer	Dr. Joachim Behrendt
Language	DE/EN
Cycle	SoSe
Content	See interlocking course
Literature	See interlocking course

Course L0357: Advanced Wastewater Treatment	
Typ	Lecture
Hrs/wk	2
CP	2
Workload in Hours	Independent Study Time 32, Study Time in Lecture 28
Lecturer	Dr. Joachim Behrendt
Language	EN
Cycle	SoSe
Content	Survey on advanced wastewater treatment reuse of reclaimed municipal wastewater Precipitation Flocculation Depth filtration Membrane Processes Activated carbon adsorption Ozonation "Advanced Oxidation Processes" Disinfection
Literature	Metcalf & Eddy, Wastewater Engineering: Treatment and Reuse, McGraw-Hill, Boston 2003 Wassertechnologie, H.H. Hahn, Springer-Verlag, Berlin 1987 Membranverfahren: Grundlagen der Modul- und Anlagenauslegung, T. Melin und R. Rautenbach, Springer-Verlag, Berlin 2007 Trinkwasserdesinfektion: Grundlagen, Verfahren, Anlagen, Geräte, Mikrobiologie, Chlorung, Ozonung, UV-Bestrahlung, Membranfiltration, Qualitätssicherung, W. Roeske, Oldenbourg-Verlag, München 2006 Organische Problemstoffe in Abwässern, H. Gulyas, GFEU, Hamburg 2003

Course L0358: Advanced Wastewater Treatment	
Typ	Recitation Section (large)
Hrs/wk	1
CP	1
Workload in Hours	Independent Study Time 16, Study Time in Lecture 14
Lecturer	Dr. Joachim Behrendt
Language	EN
Cycle	SoSe
Content	Aggregate organic compounds (sum parameters) Industrial wastewater Processes for industrial wastewater treatment Precipitation Flocculation Activated carbon adsorption Recalcitrant organic compounds
Literature	Metcalf & Eddy, Wastewater Engineering: Treatment and Reuse, McGraw-Hill, Boston 2003 Wassertechnologie, H.H. Hahn, Springer-Verlag, Berlin 1987 Membranverfahren: Grundlagen der Modul- und Anlagenauslegung, T. Melin und R. Rautenbach, Springer-Verlag, Berlin 2007 Trinkwasserdesinfektion: Grundlagen, Verfahren, Anlagen, Geräte, Mikrobiologie, Chlorung, Ozonung, UV-Bestrahlung, Membranfiltration, Qualitätssicherung, W. Roeske, Oldenbourg-Verlag, München 2006 Organische Problemstoffe in Abwässern, H. Gulyas, GFEU, Hamburg 2003

Module M1724: Smart Monitoring				
Courses				
Title	Typ		Hrs/wk	CP
Smart Monitoring (L2762)	Integrated Lecture		2	2
Smart Monitoring (L2763)	Recitation Section (small)		2	4
Module Responsible	Prof. Kay Smarsly			
Admission Requirements	None			
Recommended Previous Knowledge	Basic knowledge or interest in object-oriented modeling, programming, and sensor technologies are helpful. Interest in modern research and teaching areas, such as Internet of Things, Industry 4.0 and cyber-physical systems, as well as the will to deepen skills of scientific working, are required. Basic knowledge in scientific writing and good English skills.			
Educational Objectives	After taking part successfully, students have reached the following learning results			
Professional Competence				
<i>Knowledge</i>	The students will become familiar with the principles and practices of smart monitoring. The students will be able to design decentralized smart systems to be applied for continuous (remote) monitoring of systems in the built and in the natural environment. In addition, the students will learn to design and to implement intelligent sensor systems using state-of-the-art data analysis techniques, modern software design concepts, and embedded computing methodologies. Besides lectures, project work is also part of this module, which will be conducted throughout the semester and will contribute to the grade. In small groups, the students will design smart monitoring systems that integrate a number of "intelligent" sensors to be implemented by the students. Specific focus will be put on the application of machine learning techniques. The smart monitoring systems will be mounted on real-world (built or natural) systems, such as bridges or slopes, or on scaled lab structures for validation purposes. The outcome of every group will be documented in a paper. All students of this module will "automatically" participate with their smart monitoring system in the annual "Smart Monitoring" competition. The written papers and oral examinations form the final grades. The module will be taught in English. Limited enrollment.			
<i>Skills</i>	The students will gain insights into operating state-of-the-art smart sensor systems, used for monitoring a wide range of physical processes relevant to engineering, such as environmental, structural, or comfort monitoring. The students will be capable of devising monitoring strategies of physical processes as part of group projects, tailored to their knowledge backgrounds, and to implement the strategies in smart wireless sensor nodes, using embedded computing and programming. Finally, the students will be able to document the findings of their projects in short reports.			
Personal Competence				
<i>Social Competence</i>	The students will be able to work in groups, share parts of the work for their projects, and develop communication skills, towards achieving the common project goals.			
<i>Autonomy</i>	The students will be able to gain a solid basis on approaching and solving problems in engineering, as well as on documenting results, through their involvement in their monitoring group projects.			
Workload in Hours	Independent Study Time 124, Study Time in Lecture 56			
Credit points	6			
Course achievement	None			
Examination	Written elaboration			
Examination duration and scale	10 pages of work with 15-minute oral presentation			
Assignment for the Following Curricula	Civil Engineering: Specialisation Water and Traffic: Elective Compulsory Civil Engineering: Specialisation Geotechnical Engineering: Elective Compulsory Civil Engineering: Specialisation Coastal Engineering: Elective Compulsory Civil Engineering: Specialisation Structural Engineering: Elective Compulsory Computer Science: Specialisation II: Intelligence Engineering: Elective Compulsory Environmental Engineering: Specialisation Energy and Resources: Elective Compulsory Environmental Engineering: Specialisation Environment and Climate: Elective Compulsory Environmental Engineering: Specialisation Water Quality and Water Engineering: Elective Compulsory Mechatronics: Technical Complementary Course: Elective Compulsory Mechatronics: Core Qualification: Elective Compulsory Theoretical Mechanical Engineering: Specialisation Robotics and Computer Science: Elective Compulsory Water and Environmental Engineering: Specialisation Cities: Elective Compulsory Water and Environmental Engineering: Specialisation Environment: Elective Compulsory Water and Environmental Engineering: Specialisation Water: Elective Compulsory			

Course L2762: Smart Monitoring	
Typ	Integrated Lecture
Hrs/wk	2
CP	2
Workload in Hours	Independent Study Time 32, Study Time in Lecture 28
Lecturer	Prof. Kay Smarsly
Language	EN
Cycle	SoSe
Content	In this course, principles of smart monitoring will be taught, focusing on modern concepts of data acquisition, data storage, and data analysis. Also, fundamentals of intelligent sensors and embedded computing will be illuminated. Autonomous software and decentralized data processing are further crucial parts of the course, including concepts of the Internet of Things, Industry 4.0 and cyber-physical systems. Furthermore, measuring principles, data acquisition systems, data management and data analysis algorithms will be discussed. Besides the theoretical background, numerous practical examples will be shown to demonstrate how smart monitoring may advantageously be used for assessing the condition of systems in the built or natural environment.
Literature	The course contents couples different fields, such as signal processing, sensing technologies, data analytics, environmental engineering, civil engineering, artificial intelligence, database systems, and many more. The basics will be taught in this course. However, specific literature that covers all these topics does not exist. Instead, literature will be referenced in the lectures, all of which are papers that are freely available online.

Course L2763: Smart Monitoring	
Typ	Recitation Section (small)
Hrs/wk	2
CP	4
Workload in Hours	Independent Study Time 92, Study Time in Lecture 28
Lecturer	Prof. Kay Smarsly
Language	EN
Cycle	SoSe
Content	The contents of the exercises are based on the lecture contents. In addition to the exercises, project work will be conducted throughout the semester, which will consume the majority of the workload. As part of the project work, students will design smart monitoring systems that will be tested in the laboratory or in the field. As mentioned in the module description, the students will participate in the "Smart Monitoring" competition, hosted annually by the Institute of Digital and Autonomous Construction. Students are encouraged to contribute their own ideas. The tools required to implement the smart monitoring systems will be taught in the group exercises as well as through external sources, such as video tutorials and literature.
Literature	The course contents couples different fields, such as signal processing, sensing technologies, data analytics, environmental engineering, civil engineering, artificial intelligence, database systems, and many more. The basics will be taught in this course. However, specific literature that covers all these topics does not exist. Instead, literature will be referenced in the lectures, all of which are papers that are freely available online.

Module M0858: Coastal Hydraulic Engineering I			
Courses			
Title	Typ	Hrs/wk	CP
Basics of Coastal Engineering (L0807)	Lecture	3	4
Basics of Coastal Engineering (L1413)	Project-/problem-based Learning	1	2
Module Responsible	Prof. Peter Fröhle		
Admission Requirements	None		
Recommended Previous Knowledge	Basics of hydraulic engineering, hydrology and hydromechanics		
Educational Objectives	After taking part successfully, students have reached the following learning results		
Professional Competence	<i>Knowledge</i> The students are able to define and explain the basic concepts of coastal engineering and port engineering. They are able to apply the concepts to selected practical problems of coastal engineering. Students can define and determine the basics for design and dimensioning of coastal engineering constructions. <i>Skills</i> The students are capable to apply basic design approaches to selected and pre-defined design tasks in coastal engineering. <i>Social Competence</i> The students are able to deploy their gained knowledge in applied problems such as the design of coastal protection structures. Additionally, they will be able to work in team with engineers of other disciplines, for instance designing of coastal breakwaters. <i>Autonomy</i> The students will be able to independently extend their knowledge and apply it to new problems.		
Workload in Hours	Independent Study Time 124, Study Time in Lecture 56		
Credit points	6		
Course achievement	None		
Examination	Written exam		
Examination duration and scale	The duration of the examination is 2 hours. The examination includes tasks with respect to the general understanding of the lecture contents and calculations tasks.		
Assignment for the Following Curricula	Civil Engineering: Specialisation Coastal Engineering: Compulsory Civil Engineering: Specialisation Geotechnical Engineering: Compulsory Civil Engineering: Specialisation Structural Engineering: Elective Compulsory Environmental Engineering: Specialisation Environment and Climate: Elective Compulsory Environmental Engineering: Specialisation Water Quality and Water Engineering: Elective Compulsory International Management and Engineering: Specialisation II. Civil Engineering: Elective Compulsory Water and Environmental Engineering: Specialisation Environment: Elective Compulsory Water and Environmental Engineering: Specialisation Water: Elective Compulsory		

Course L0807: Basics of Coastal Engineering	
Typ	Lecture
Hrs/wk	3
CP	4
Workload in Hours	Independent Study Time 78, Study Time in Lecture 42
Lecturer	Prof. Peter Fröhle
Language	EN
Cycle	SoSe
Content	- Basics of planning and design - Water levels - Currents - Waves - Ice - Planning and Design in Coastal Engineering - Functional and constructional design - Determination of design parameters - Design-approaches - Filter - Rubble mound constructions - Piles - Vertical constructions
Literature	Coastal Engineering Manual, CEM Vorlesungsumdruck

Course L1413: Basics of Coastal Engineering	
Typ	Project-/problem-based Learning
Hrs/wk	1
CP	2
Workload in Hours	Independent Study Time 46, Study Time in Lecture 14
Lecturer	Prof. Peter Fröhle
Language	EN
Cycle	SoSe
Content	See interlocking course
Literature	See interlocking course

Module M1898: Study Work Water Quality and Water Engineering				
Courses				
Title	Typ		Hrs/wk	CP
Module Responsible	Dozenten des Studiengangs			
Admission Requirements	None			
Recommended Previous Knowledge				
Educational Objectives	After taking part successfully, students have reached the following learning results			
Professional Competence <i>Knowledge</i>	Students are able to demonstrate their detailed knowledge in a field of water and environmental engineering. The students are qualified to project water technology and environmental protection-oriented projects and to independently define research tasks for the theoretical and experimental investigation of environmental problems and water management issues. They are able to give examples of the state of development and application and to discuss these critically, taking into account current problems and framework conditions in science and society. The students are able to independently define a solution strategy for a basic, application-oriented or practical problem from the field of water and environmental engineering and to outline individual solution approaches. They can proceed in a theory-oriented manner and include current safety, ecological, ethical and economic aspects according to the state of the art in science and related social discussions. They can use the scientific working techniques they have chosen for their own project work, they can present them in detail and critically discuss them.			
<i>Skills</i>	Students are able to independently select methodological approaches for project work and justify this selection in terms of content. They can explain how they relate approaches or methods to the specific field of application in a solution-oriented manner and adapt them to the application context. They can outline the main points and further developments that go beyond the project.			
Personal Competence <i>Social Competence</i>	Students are able to prepare the relevance and cut of their project task, the work steps and sub-problems for discussion and debate in larger groups, guide the discussions and give feedback to colleagues on their projects.			
<i>Autonomy</i>	The students are able to independently plan and document the work steps and processes necessary to complete the coursework, taking into account specified deadlines. This includes being able to obtain current scientific information in a goal-oriented manner. Furthermore, they are able to obtain feedback on the progress of work from experts in the field in order to achieve high-quality work results based on the state of the art in science and technology.			
Workload in Hours	Independent Study Time 360, Study Time in Lecture 0			
Credit points	12			
Course achievement	None			
Examination	Study work			
Examination duration and scale	depending on task			
Assignment for the Following Curricula	Environmental Engineering: Specialisation Water Quality and Water Engineering: Compulsory			

Module M0802: Membrane Technology								
Courses								
Title		Typ	Hrs/wk	CP				
Membrane Technology (L0399)		Lecture	2	3				
Membrane Technology (L0400)		Recitation Section (small)	1	2				
Membrane Technology (L0401)		Practical Course	1	1				
Module Responsible	Prof. Mathias Ernst							
Admission Requirements	None							
Recommended Previous Knowledge	Basic knowledge of water chemistry. Knowledge of the core processes involved in water, gas and steam treatment							
Educational Objectives	After taking part successfully, students have reached the following learning results							
Professional Competence	Knowledge Students will be able to rank the technical applications of industrially important membrane processes. They will be able to explain the different driving forces behind existing membrane separation processes. Students will be able to name materials used in membrane filtration and their advantages and disadvantages. Students will be able to explain the key differences in the use of membranes in water, other liquid media, gases and in liquid/gas mixtures. Skills Students will be able to prepare mathematical equations for material transport in porous and solution-diffusion membranes and calculate key parameters in the membrane separation process. They will be able to handle technical membrane processes using available boundary data and provide recommendations for the sequence of different treatment processes. Through their own experiments, students will be able to classify the separation efficiency, filtration characteristics and application of different membrane materials. Students will be able to characterise the formation of the fouling layer in different waters and apply technical measures to control this. Personal Competence Social Competence Students will be able to work in diverse teams on tasks in the field of membrane technology. They will be able to make decisions within their group on laboratory experiments to be undertaken jointly and present these to others. Autonomy Students will be in a position to solve homework on the topic of membrane technology independently. They will be capable of finding creative solutions to technical questions.							
Workload in Hours					Independent Study Time 124, Study Time in Lecture 56			
Credit points					6			
Course achievement					None			
Examination	Written exam							
Examination duration and scale	90 min							
Assignment for the Following Curricula	Civil Engineering: Specialisation Water and Traffic: Elective Compulsory Bioprocess Engineering: Specialisation A - General Bioprocess Engineering: Elective Compulsory Bioprocess Engineering: Specialisation B - Industrial Bioprocess Engineering: Elective Compulsory Chemical and Bioprocess Engineering: Specialisation General Process Engineering: Elective Compulsory Chemical and Bioprocess Engineering: Specialisation Chemical Process Engineering: Elective Compulsory Chemical and Bioprocess Engineering: Technical Complementary Course: Elective Compulsory Chemical and Bioprocess Engineering: Technical Complementary Course: Elective Compulsory Environmental Engineering: Specialisation Water Quality and Water Engineering: Elective Compulsory Process Engineering: Specialisation Process Engineering: Elective Compulsory Process Engineering: Specialisation Environmental Process Engineering: Elective Compulsory Water and Environmental Engineering: Specialisation Water: Elective Compulsory Water and Environmental Engineering: Specialisation Environment: Elective Compulsory Water and Environmental Engineering: Specialisation Cities: Elective Compulsory							

Course L0399: Membrane Technology	
Typ	Lecture
Hrs/wk	2
CP	3
Workload in Hours	Independent Study Time 62, Study Time in Lecture 28
Lecturer	Prof. Mathias Ernst
Language	EN
Cycle	WiSe
Content	The lecture on membrane technology supply provides students with a broad understanding of existing membrane treatment processes, encompassing pressure driven membrane processes, membrane application in electrodialysis, pervaporation as well as membrane distillation. The lectures main focus is the industrial production of drinking water like particle separation or desalination; however gas separation processes as well as specific wastewater oriented applications such as membrane bioreactor systems will be discussed as well. Initially, basics in low pressure and high pressure membrane applications are presented (microfiltration, ultrafiltration, nanofiltration, reverse osmosis). Students learn about essential water quality parameter, transport equations and key parameter for pore membrane as well as solution diffusion membrane systems. The lecture sets a specific focus on fouling and scaling issues and provides knowledge on methods how to tackle with these phenomena in real water treatment application. A further part of the lecture deals with the character and manufacturing of different membrane materials and the characterization of membrane material by simple methods and advanced analysis. The functions, advantages and drawbacks of different membrane housings and modules are explained. Students learn how an industrial membrane application is designed in the succession of treatment steps like pre-treatment, water conditioning, membrane integration and post-treatment of water. Besides theory, the students will be provided with knowledge on membrane demo-site examples and insights in industrial practice.
Literature	• T. Melin, R. Rautenbach: Membranverfahren: Grundlagen der Modul- und Anlagenauslegung (2., erweiterte Auflage), Springer-Verlag, Berlin 2004. • Marcel Mulder, Basic Principles of Membrane Technology, Kluwer Academic Publishers, Dordrecht, The Netherlands • Richard W. Baker, Membrane Technology and Applications, Second Edition, John Wiley & Sons, Ltd., 2004

Course L0400: Membrane Technology	
Typ	Recitation Section (small)
Hrs/wk	1
CP	2
Workload in Hours	Independent Study Time 46, Study Time in Lecture 14
Lecturer	Prof. Mathias Ernst
Language	EN
Cycle	WiSe
Content	See interlocking course
Literature	See interlocking course

Course L0401: Membrane Technology	
Typ	Practical Course
Hrs/wk	1
CP	1
Workload in Hours	Independent Study Time 16, Study Time in Lecture 14
Lecturer	Prof. Mathias Ernst
Language	EN
Cycle	WiSe
Content	See interlocking course
Literature	See interlocking course

Module M0822: Process Modeling in Water Technology				
Courses				
Title	Typ		Hrs/wk	CP
Process Modelling of Wastewater Treatment (L0522)	Project-/problem-based Learning		2	3
Process Modeling in Drinking Water Treatment (L0314)	Project-/problem-based Learning		2	3
Module Responsible	Dr. Klaus Johannsen			
Admission Requirements	None			
Recommended Previous Knowledge	Knowledge of the most important processes in drinking water and waste water treatment.			
Educational Objectives	After taking part successfully, students have reached the following learning results			
Professional Competence	<i>Knowledge</i> Students are able to explain selected processes of drinking water and waste water treatment in detail. They are able to explain basics as well as possibilities and limitations of dynamic modeling. <i>Skills</i> Students are able to use the most important features Modelica offers. They are able to transpose selected processes in drinking water and waste water treatment into a mathematical model in Modelica with respect to equilibrium, kinetics and mass balances. They are able to set up and apply models and assess their possibilities and limitations. <i>Social Competence</i> Students are able to solve problems and document solutions in a group with members of different technical background. They are able to give appropriate feedback and can work constructively with feedback concerning their work. <i>Autonomy</i> Students are able to define a problem, gain the required knowledge and set up a model.			
Workload in Hours	Independent Study Time 124, Study Time in Lecture 56			
Credit points	6			
Course achievement	None			
Examination	Oral exam			
Examination duration and scale	30 min			
Assignment for the Following Curricula	Civil Engineering: Specialisation Water and Traffic: Elective Compulsory Chemical and Bioprocess Engineering: Technical Complementary Course: Elective Compulsory Chemical and Bioprocess Engineering: Technical Complementary Course: Elective Compulsory Environmental Engineering: Specialisation Water Quality and Water Engineering: Elective Compulsory Process Engineering: Specialisation Environmental Process Engineering: Elective Compulsory Process Engineering: Specialisation Process Engineering: Elective Compulsory Water and Environmental Engineering: Specialisation Water: Elective Compulsory Water and Environmental Engineering: Specialisation Environment: Elective Compulsory Water and Environmental Engineering: Specialisation Cities: Elective Compulsory			

Course L0522: Process Modelling of Wastewater Treatment	
Typ	Project-/problem-based Learning
Hrs/wk	2
CP	3
Workload in Hours	Independent Study Time 62, Study Time in Lecture 28
Lecturer	Dr. Joachim Behrendt
Language	DE/EN
Cycle	WiSe
Content	Mass and energy balances Tracer modelling Activated Sludge Model Wastewater Treatment Plant Modelling (continuously and SBR) Sludge Treatment (ADM, aerobic autothermal) Biofilm Modelling
Literature	Henze, Mogens (Seminar on Activated Sludge Modelling, ; Kollekolle Seminar on Activated Sludge Modelling, ;) Activated sludge modelling : processes in theory and practice ; selected proceedings of the 5th Kollekolle Seminar on Activated Sludge Modelling, held in Kollekolle, Denmark, 10 - 12 September 2001 ISBN: 1843394146 [London] : IWA Publ., 2002 TUB_HH_Katalog Henze, Mogens Activated sludge models ASM1, ASM2, ASM2d and ASM3 ISBN: 1900222248 London : IWA Publ., 2002 TUB_HH_Katalog Henze, Mogens Wastewater treatment : biological and chemical processes ISBN: 3540422285 (Pp.) Berlin [u.a.] : Springer, 2002 TUB_HH_Katalog Wiesmann, Udo (Choi, In Su; Dombrowski, Eva-Maria;) Fundamentals of biological wastewater treatment ISBN: 3527312196 (Gb.) URL: http://deposit.ddb.de/cgi-bin/dokserv?id=2774611&prov=M&dok_var=1&dok_ext=htm Weinheim : WILEY-VCH, 2007 TUB_HH_Katalog

Course L0314: Process Modeling in Drinking Water Treatment	
Typ	Project-/problem-based Learning
Hrs/wk	2
CP	3
Workload in Hours	Independent Study Time 62, Study Time in Lecture 28
Lecturer	Dr. Klaus Johannsen
Language	EN
Cycle	WiSe
Content	In this course selected drinking water treatment processes (e.g. aeration or activated carbon adsorption) are modeled dynamically using the programming language Modelica, that is increasingly used in industry. In this course OpenModelica is used, an free access frontend of the programming language Modelica. In the beginning of the course the use of OpenModelica is explained by means of simple examples. Together required elements and structure of the model are developed. The implementation in OpenModelica and the application of the model is done individually or in groups respectively. Students get feedback and can gain extra points for the exam.
Literature	OpenModelica: https://openmodelica.org/index.php/download/download-windows OpenModelica - Modelica Tutorial: https://openmodelica.org/index.php/useresresources/userdocumentation OpenModelica - Users Guide: https://openmodelica.org/index.php/useresresources/userdocumentation Peter Fritzson: Principles of Object-Oriented Modeling and Simulation with Modelica 2.1, Wiley-IEEE Press, ISBN 0-471-471631. MHW (rev. by Crittenden, J. et al.): Water treatment principles and design. John Wiley & Sons, Hoboken, 2005. Stumm, W., Morgan, J.J.: Aquatic chemistry. John Wiley & Sons, New York, 1996. DVGW (Hrsg.): Wasseraufbereitung - Grundlagen und Verfahren. Oldenbourg Industrie Verlag, München, 2004.

Module M0859: Coastal Hydraulic Engineering II			
Courses			
Title	Typ	Hrs/wk	CP
Coastal- and Flood Protection (L0808)	Lecture	2	3
Coastal- and Flood Protection (L1415)	Project-/problem-based Learning	1	1
Maintenance and Defence of Flood Protection Structures (L1411)	Lecture	2	2
Module Responsible	Prof. Peter Fröhle		
Admission Requirements	None		
Recommended Previous Knowledge	Coastal Engineering I		
Educational Objectives	After taking part successfully, students have reached the following learning results		
Professional Competence	The students have the capability to define and explain in detail the important aspects of erosion protection and flood protection and are able to apply the aspects to practical coastal protection problems. They are able to design and dimension important coastal protection measures from the functional and from the constructional point of view. The students are able to select design approaches for the functional and constructional design of erosion and flood protection measures and apply these approaches to practical design tasks. The students are able to deploy their gained knowledge in applied problems such as the functional and constructive design of coastal and flood protection structures. Additionally, they will be able to work in team with engineers of other disciplines. The students will be able to independently extend their knowledge and apply it to new problems.		
Knowledge			
Skills			
Personal Competence			
Social Competence			
Autonomy			
Workload in Hours	Independent Study Time 110, Study Time in Lecture 70		
Credit points	6		
Course achievement	None		
Examination	Written exam		
Examination duration and scale	The duration of the examination is 130 min. The examination includes tasks with respect to the general understanding of the lecture contents and calculations tasks.		
Assignment for the Following Curricula	Civil Engineering: Specialisation Coastal Engineering: Compulsory Civil Engineering: Specialisation Geotechnical Engineering: Elective Compulsory Civil Engineering: Specialisation Structural Engineering: Elective Compulsory Environmental Engineering: Specialisation Environment and Climate: Elective Compulsory Environmental Engineering: Specialisation Water Quality and Water Engineering: Elective Compulsory Water and Environmental Engineering: Specialisation Environment: Elective Compulsory Water and Environmental Engineering: Specialisation Water: Elective Compulsory		

Course L0808: Coastal- and Flood Protection	
Typ	Lecture
Hrs/wk	2
CP	3
Workload in Hours	Independent Study Time 62, Study Time in Lecture 28
Lecturer	Prof. Peter Fröhle
Language	EN
Cycle	WiSe
Content	Protection of sandy coasts • Sediment transport • Morphology • Technical solution for the protection of sandy coasts ◦ Construction in direction of the coast ◦ Constructions perpendicular to the coast ◦ Other Concept • Calculation approaches and numerical models Flood Protection • Classification of constructions / measures • Dikes • Dunes • Foreland - constructions • Flood-Protection Walls • Drainage of the hinterland
Literature	Vorlesungsumdruck Coastal Engineering Manual CEM

Course L1415: Coastal- and Flood Protection	
Typ	Project-/problem-based Learning
Hrs/wk	1
CP	1
Workload in Hours	Independent Study Time 16, Study Time in Lecture 14
Lecturer	Prof. Peter Fröhle
Language	EN
Cycle	WiSe
Content	See interlocking course
Literature	See interlocking course

Course L1411: Maintenance and Defence of Flood Protection Structures	
Typ	Lecture
Hrs/wk	2
CP	2
Workload in Hours	Independent Study Time 32, Study Time in Lecture 28
Lecturer	Dr. Olaf Müller
Language	EN
Cycle	WiSe
Content	• Dike protection • Maintenance of flood protection measures
Literature	Vorlesungsumdruck

Module M2156: Water Protection				
Courses				
Title	Typ		Hrs/wk	CP
Water Protection (L3459)	Integrated Lecture		6	6
Module Responsible	Prof. Simon Michael Papalexiou			
Admission Requirements	None			
Recommended Previous Knowledge	• Basic knowledge in water management; • Good knowledge in urban drainage; • Good knowledge of wastewater treatment techniques; • Good knowledge of pollutants (e.g. COD, BOD, TS, N, P) and their properties;			
Educational Objectives	After taking part successfully, students have reached the following learning results			
Professional Competence	<i>Knowledge</i> The students can describe the basic principles of the regulatory framework related to the international and European water sector. They can explain limnological processes, substance cycles and water morphology in detail. They are able to assess complex problems related to water protection, such as ecosystem service and wastewater treatment with a special focus on innovative solutions, remediation measures as well as conceptual approaches. <i>Skills</i> Students can accurately assess current problems and situations in a country-specific or local context. They can suggest concrete actions to contribute to the planning of tomorrow's urban water cycle. Furthermore, they can suggest appropriate technical, administrative and legislative solutions to solve these problems. Personal Competence <i>Social Competence</i> The students can work together in international groups. <i>Autonomy</i> Students are able to organize their work flow to prepare presentations and discussions. They can acquire appropriate knowledge by making enquiries independently.			
Workload in Hours	Independent Study Time 96, Study Time in Lecture 84			
Credit points	6			
Course achievement	Compulsory	Bonus	Form	Description
	Yes	20 %	Presentation	10-minütige Präsentation von Arbeitsergebnissen
Examination	Written exam			
Examination duration and scale	150 minutes			
Assignment for the Following Curricula	Civil Engineering: Specialisation Coastal Engineering: Elective Compulsory Civil Engineering: Specialisation Geotechnical Engineering: Elective Compulsory Civil Engineering: Specialisation Structural Engineering: Elective Compulsory Civil Engineering: Specialisation Water and Traffic: Elective Compulsory Environmental Engineering: Specialisation Water Quality and Water Engineering: Elective Compulsory International Management and Engineering: Specialisation II. Civil Engineering: Elective Compulsory Water and Environmental Engineering: Specialisation Cities: Elective Compulsory Water and Environmental Engineering: Specialisation Environment: Compulsory Water and Environmental Engineering: Specialisation Water: Elective Compulsory			

Course L3459: Water Protection	
Typ	Integrated Lecture
Hrs/wk	6
CP	6
Workload in Hours	Independent Study Time 96, Study Time in Lecture 84
Lecturer	Prof. Simon Michael Papalexiou
Language	EN
Cycle	WiSe
Content	
Literature	

Thesis

Module M-002: Master Thesis

Courses

Title	Typ	Hrs/wk	CP
Module Responsible	Professoren der TUHH		
Admission Requirements	According to General Regulations §21 (1): At least 60 credit points have to be achieved in study programme. The examinations board decides on exceptions.		
Recommended Previous Knowledge			
Educational Objectives	After taking part successfully, students have reached the following learning results		
Professional Competence <i>Knowledge</i>	- The students can use specialized knowledge (facts, theories, and methods) of their subject competently on specialized issues. - The students can explain in depth the relevant approaches and terminologies in one or more areas of their subject, describing current developments and taking up a critical position on them. - The students can place a research task in their subject area in its context and describe and critically assess the state of research.		
Skills	The students are able: - To select, apply and, if necessary, develop further methods that are suitable for solving the specialized problem in question. - To apply knowledge they have acquired and methods they have learnt in the course of their studies to complex and/or incompletely defined problems in a solution-oriented way. - To develop new scientific findings in their subject area and subject them to a critical assessment.		
Personal Competence <i>Social Competence</i>	Students can - Both in writing and orally outline a scientific issue for an expert audience accurately, understandably and in a structured way. - Deal with issues competently in an expert discussion and answer them in a manner that is appropriate to the addressees while upholding their own assessments and viewpoints convincingly.		
<i>Autonomy</i>	Students are able: - To structure a project of their own in work packages and to work them off accordingly. - To work their way in depth into a largely unknown subject and to access the information required for them to do so. - To apply the techniques of scientific work comprehensively in research of their own.		
Workload in Hours	Independent Study Time 900, Study Time in Lecture 0		
Credit points	30		
Course achievement	None		
Examination	Thesis		
Examination duration and scale	According to General Regulations		
Assignment for the Following Curricula	Civil Engineering: Thesis: Compulsory Bioprocess Engineering: Thesis: Compulsory Chemical and Bioprocess Engineering: Thesis: Compulsory Chemical and Bioprocess Engineering: Thesis: Compulsory Computational Engineering: Thesis: Compulsory Computer Science: Thesis: Compulsory Data Science: Thesis: Compulsory Electrical Engineering and Information Technology: Thesis: Compulsory Electrical Engineering: Thesis: Compulsory Energy Systems: Thesis: Compulsory Environmental Engineering: Thesis: Compulsory Aircraft Systems Engineering: Thesis: Compulsory Global Innovation Management: Thesis: Compulsory Computer Science in Engineering: Thesis: Compulsory Information and Communication Systems: Thesis: Compulsory Interdisciplinary Mathematics: Thesis: Compulsory International Production Management: Thesis: Compulsory International Management and Engineering: Thesis: Compulsory		

Joint European Master in Environmental Studies - Cities and Sustainability: Thesis: Compulsory
Logistics, Infrastructure and Mobility: Thesis: Compulsory
Aeronautics: Thesis: Compulsory
Mechanical Engineering - Product Development and Production: Thesis: Compulsory
Materials Science and Engineering: Thesis: Compulsory
Materials Science: Thesis: Compulsory
Mechanical Engineering and Management: Thesis: Compulsory
Mechatronics: Thesis: Compulsory
Biomedical Engineering: Thesis: Compulsory
Microelectronics and Microsystems: Thesis: Compulsory
Product Development, Materials and Production: Thesis: Compulsory
Renewable Energies: Thesis: Compulsory
Naval Architecture and Ocean Engineering: Thesis: Compulsory
Naval Architecture and Ocean Engineering: Thesis: Compulsory
Ship and Offshore Technology: Thesis: Compulsory
Theoretical Mechanical Engineering: Thesis: Compulsory
Process Engineering: Thesis: Compulsory
Water and Environmental Engineering: Thesis: Compulsory
Certification in Engineering & Advisory in Aviation: Thesis: Compulsory